

MSSP Results Analyzer – Energy Community (EC)

Energy System Market Analysis

Variables:

Group	Variables
Flexible Demand	var_fd , var_afd_p , var_afd_m
Generation	pPV , pW
BESS	cV_dV (net charge rate), idV (state), socV (SoC)
Day-Ahead	eDA_p , eDA_m , ieDA_p , ieDA_m
Reserves	rU , rD , rU_B , rD_B , rU_FD , rD_FD
Imbalance	eIM , eIM1 , eIM2 , eIM3 , pIB_m , pIB_p

Data format expected: one CSV per variable — rows = time steps, columns = scenarios.

CI bands show P10–P90 across scenarios unless otherwise noted.

0 · Imports & Setup

```
In [1]: import os, glob, warnings
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
from matplotlib.ticker import MaxNLocator
from scipy import stats

warnings.filterwarnings("ignore")
%matplotlib inline
plt.rcParams.update({
    "figure.dpi": 120,
    "axes.spines.top": False,
    "axes.spines.right": False,
    "font.size": 10,
})
print("Libraries loaded ✓")
```

Libraries loaded ✓

1 · Configuration

Edit the block below to point to your data. Three loading modes are supported:

- **Option A** - one CSV per variable (rows = timesteps, cols = scenarios) → set `data_dir`
- **Option B** - single wide CSV with `(variable, scenario)` MultiIndex columns → set `single_csv`
- **Option C** - HDF5 store with variable names as keys → set `hdf5_file`

If no files are found the script falls back to **synthetic data** so every cell runs out-of-the-box.

```
In [2]: CONFIG = dict(  
    # — Data location —  
    data_dir = "../mmobec-pyomo/results/FTC_20260201_20260228_c10_sc15_15m",  
    file_ext = ".txt", # Option A: folder with one TXT per variable  
    single_csv = None, # Option B: path to wide CSV  
    hdf5_file = None, # Option C: path to HDF5  
  
    # — Time axis —  
    n_periods = 24, # number of time steps (e.g. 24 h)  
    period_label = "Hour", # x-axis label  
  
    # — CI percentiles —  
    ci_low = 10,  
    ci_high = 90,  
  
    # — Save figures —  
    save_figs = False, # set True to export PNGs  
    fig_dir = "figures/",  
    fig_format = "png",  
    dpi = 150,  
)  
  
UNITS = {  
    "var_fd": "MW", "var_afd_p": "MW", "var_afd_m": "MW",  
    "pPV": "MW", "pW": "MW",  
    "cV_dV": "MW", "idV": "-", "socV": "MWh",  
    "eDA_p": "MWh", "eDA_m": "MWh", "ieDA_p": "MWh", "ieDA_m": "MWh",  
    "rU": "MW", "rD": "MW",  
    "rU_B": "MW", "rD_B": "MW", "rU_FD": "MW", "rD_FD": "MW",  
    "eIM": "MWh", "eIM_1": "MWh", "eIM_2": "MWh", "eIM_3": "MWh", "pIB_m": "MW",  
}  
  
COLORS = dict(  
    median="#1a1a2e", ci_fill="#4a90d9",  
    pos="#27ae60", neg="#e74c3c",  
    bess="#8e44ad", pv="#f39c12",  
    wind="#2980b9", reserve="#16a085",  
    imbal="#c0392b", fd="#d35400",  
    da="#2c3e50",  
)  
print("Configuration set ✓")
```

2 · Data Loader

```
In [3]: def _synthetic_data(cfg):
    """Fallback synthetic data – replace with real loader if needed."""
    rng = np.random.default_rng(42)
    T, S = cfg["n_periods"], 50
    t = np.linspace(0, 2*np.pi, T)
    scen = [f"s{i:03d}" for i in range(S)]
    data = {}

    def _df(base, noise=0.15, lo=None, hi=None):
        mat = base[:,None] + rng.normal(0, noise*(np.abs(base).mean()+0.01),
        if lo is not None: mat = np.clip(mat, lo, None)
        if hi is not None: mat = np.clip(mat, None, hi)
        return pd.DataFrame(mat, columns=scen)

    solar = np.maximum(0, np.sin(t - np.pi/3))
    wind = 0.4 + 0.3*np.sin(2*t + 0.5)
    dem = 1.0 + 0.4*np.sin(t - np.pi/2)
    sf_b = np.where(rng.random(T) > 0.75, rng.uniform(0,0.3,T), 0)

    data["pPV"] = _df(solar*2.0, lo=0)
    data["pW"] = _df(wind*1.5, lo=0)
    data["socV"] = _df(0.5+0.3*np.sin(t), noise=0.05, lo=0, hi=1.0)
    data["cV_dV"] = _df(0.3*np.sin(t+1.0), noise=0.2)
    data["idV"] = pd.DataFrame((rng.random((T,S))>0.5).astype(float), c
    data["var_fd"] = _df(dem, lo=0)
    data["var_afd_p"] = _df(np.abs(0.1*np.sin(t)), lo=0)
    data["var_afd_m"] = _df(np.abs(0.1*np.cos(t)), lo=0)
    eDA = dem*0.9
    data["eDA_p"] = _df(eDA, lo=0)
    data["eDA_m"] = _df(eDA*0.1, lo=0)
    data["ieDA_p"] = _df(sf_b, noise=0.3, lo=0)
    data["ieDA_m"] = _df(sf_b*0.5, noise=0.3, lo=0)
    data["rU"] = _df(0.2+0.1*np.sin(t), lo=0)
    data["rD"] = _df(0.2+0.1*np.cos(t), lo=0)
    data["rU_B"] = _df(0.1*np.abs(np.sin(t)), lo=0)
    data["rD_B"] = _df(0.1*np.abs(np.cos(t)), lo=0)
    data["rU_FD"] = _df(0.1*np.abs(np.sin(t+0.5)), lo=0)
    data["rD_FD"] = _df(0.1*np.abs(np.cos(t+0.5)), lo=0)
    data["eIM"] = _df(0.05*np.sin(3*t))
    data["pIB_p"] = _df(sf_b*0.8, noise=0.4, lo=0)
    data["pIB_m"] = _df(sf_b*0.6, noise=0.4, lo=0)
    return data

def load_data(cfg):
    data = {}
    if cfg.get("hdf5_file") and os.path.exists(cfg["hdf5_file"]):
        with pd.HDFStore(cfg["hdf5_file"], mode="r") as s:
            for k in s.keys():
                data[k.lstrip("/").replace("-", "_")] = s[k]
        return data
    if cfg.get("single_csv") and os.path.exists(cfg["single_csv"]):
```

```

wide = pd.read_csv(cfg["single_csv"], header=[0,1], index_col=0)
for v in wide.columns.get_level_values(0).unique():
    data[v.replace("-", "_")] = wide[v]
return data
files = glob.glob(os.path.join(cfg["data_dir"], "**", f"*{cfg['file_ext']}"))
if not files:
    print("⚠ No result files found – using SYNTHETIC data.")
    return _synthetic_data(cfg)

# MMOBEC 15-min block files: one scenario is stored as n_periods rows.
# First row of each block starts with: scenario probability q1 q2 ... qn
BLOCK_TXT_VARS = {
    "lPIB", "lNIB", "pIB_p", "pIB_m", "lI_1", "lI_2", "lI_3",
    "eDA_p", "eDA_m", "pW", "pPV", "dV", "cV", "var_fd", "eIM",
    "eIM_1", "eIM_2", "eIM_3",
}

def _scen_name(val):
    try:
        return f"s{int(float(val)):03d}"
    except Exception:
        return str(val)

def _wide_from_matrix(values, scen_names, n_q=None):
    values = values.astype(float).T
    values.columns = scen_names
    n_t_cfg = int(cfg.get("n_periods", 24))
    if n_q is None and values.shape[0] % n_t_cfg == 0:
        n_q = values.shape[0] // n_t_cfg
    if n_q and values.shape[0] % n_q == 0:
        n_t = values.shape[0] // n_q
        values.index = pd.MultiIndex.from_tuples(
            [(t, q) for t in range(1, n_t + 1) for q in range(1, n_q + 1)],
            names=["t", "q"],
        )
    else:
        values.index = np.arange(1, values.shape[0] + 1)
    return values

def _load_block_txt(path, name):
    rows = []
    with open(path, "r", encoding="utf-8") as fh:
        for line in fh:
            line = line.strip()
            if line and not line.startswith("#"):
                rows.append(line.split())
    if not rows:
        return None

    n_t = int(cfg.get("n_periods", 24))
    i = 0
    scen_names, vectors = [], []
    while i < len(rows):
        header = rows[i]
        if len(header) < 3 or i + n_t > len(rows):
            return None

```

```

        scenario = header[0]
        first_period_values = [float(x) for x in header[2:]]
        n_q = len(first_period_values)
        vector = first_period_values[:]

        for j in range(1, n_t):
            period_values = [float(x) for x in rows[i + j]]
            if len(period_values) != n_q:
                return None
            vector.extend(period_values)

        scen_names.append(_scen_name(scenario))
        vectors.append(vector)
        i += n_t

    cfg.setdefault("_txt_subperiods", {})[name] = n_q
    return _wide_from_matrix(pd.DataFrame(vectors, index=scen_names), sc

def _load_txt(path, name):
    if name in BLOCK_TXT_VARS:
        block_df = _load_block_txt(path, name)
        if block_df is not None:
            return block_df

    df = pd.read_csv(path, sep=r"\s+", header=None, comment="#", engine=
    if df.shape[1] <= 2 or df.shape[0] < 2:
        return None
    scen_names = [_scen_name(v) for v in df.iloc[:, 0]]
    values = df.iloc[:, 2:].astype(float)
    return _wide_from_matrix(values, scen_names)

for f in sorted(files):
    base = os.path.splitext(os.path.basename(f))[0]
    if base.endswith("_params") or base in {"numscen", "profit_ec", "tim

        continue
    name = base.replace("-", "_")
    if name == "eIM_TOT":
        name = "eIM"

    df = _load_txt(f, name)
    if df is None:
        continue
    data[name] = df
return data

# — helpers —
def _t(df):
    if isinstance(df.index, pd.MultiIndex) and {"t", "q"}.issubset(df.index.
        t_vals = df.index.get_level_values("t").astype(float).to_numpy()
        q_vals = df.index.get_level_values("q").astype(float).to_numpy()
        n_q = max(1, int(df.index.get_level_values("q").nunique()))
        return t_vals + (q_vals - 1) / n_q
    return np.arange(1, df.shape[0] + 1)

def _format_hour_axis(ax, t=None):
    n_periods = int(CONFIG.get("n_periods", 24))

```

```

ticks = np.arange(1, n_periods + 1)
ax.set_xticks(ticks)
ax.set_xticklabels([str(i) for i in ticks])
ax.set_xlim(1, n_periods + 1 - 1 / max(1, int(CONFIG.get("_active_n_q",

def _operational_ts_data(data, ref_names=("socV", "cV", "pIB_p", "pPV")):
    """Keep variables that share the operational (t, q) time index."""
    ref = next((data[name] for name in ref_names if name in data), None)
    if ref is None:
        return {}
    return {
        name: df
        for name, df in data.items()
        if isinstance(df, pd.DataFrame) and df.index.equals(ref.index)
    }

def _ci(df, lo=None, hi=None):
    lo = lo or CONFIG["ci_low"]; hi = hi or CONFIG["ci_high"]
    return df.median(axis=1).values, np.percentile(df.values, lo, axis=1), np.

def _band(ax, t, df, color, label, alpha=0.18, lw=2, lo=None, hi=None):
    med, low, high = _ci(df, lo, hi)
    ax.fill_between(t, low, high, color=color, alpha=alpha, zorder=1)
    ax.plot(t, med, color=color, lw=lw, label=label, zorder=2)
    _format_hour_axis(ax, t)
    return med, low, high

def _annotate_socv_initial(ax):
    init = data_meta.get("socV_initial")
    if init is None:
        return
    init_mean = float(init.mean())
    ax.axhline(init_mean, color=COLORS["bess"], lw=1.2, ls=":", alpha=0.9,
               label=f"Initial SoC mean ({init_mean:.3f})")

def _save(fig, name):
    if CONFIG["save_figs"]:
        os.makedirs(CONFIG["fig_dir"], exist_ok=True)
        fig.savefig(os.path.join(CONFIG["fig_dir"], f"{name}.{CONFIG['fig_fc']}"
                                dpi=CONFIG["dpi"], bbox_inches="tight")

# — LOAD —
data = load_data(CONFIG)
data_meta = {}

def _trim_initial_socv(data):
    """socV is saved with one extra initial state; plots use only operational
    if "socV" not in data:
        return data
    ref_name = next((name for name in ["cV", "dV", "cV_dV", "pIB_p"] if name
    if ref_name is None:
        return data

    soc = data["socV"]

```

```

ref = data[ref_name]
if soc.shape[0] == ref.shape[0] + 1:
    data_meta["socV_initial"] = soc.iloc[0].copy()
    soc_plot = soc.iloc[1:].copy()
    if len(soc_plot) == len(ref):
        soc_plot.index = ref.index
    data["socV"] = soc_plot
    if isinstance(soc_plot.index, pd.MultiIndex) and "q" in soc_plot.index.names:
        CONFIG["_active_n_q"] = int(soc_plot.index.get_level_values("q").max() + 1)
    init_mean = data_meta["socV_initial"].mean()
    print(f"Dropped initial socV state for plots: socV now shape={data['socV'].shape}")
return data

data = _trim_initial_socv(data)
print(f"Variables loaded ({len(data)}): {sorted(data.keys())}")
for v, df in sorted(data.items()):

    print(f" {v:<14} shape={df.shape}    NaN={df.isna().sum().sum()}")

```

△ No result files found – using SYNTHETIC data.

Variables loaded (21): ['cV_dV', 'eDA_m', 'eDA_p', 'eIM', 'idV', 'ieDA_m', 'ieDA_p', 'pIB_m', 'pIB_p', 'pPV', 'pW', 'rD', 'rD_B', 'rD_FD', 'rU', 'rU_B', 'rU_FD', 'socV', 'var_afd_m', 'var_afd_p', 'var_fd']

cV_dV	shape=(24, 50)	NaN=0
eDA_m	shape=(24, 50)	NaN=0
eDA_p	shape=(24, 50)	NaN=0
eIM	shape=(24, 50)	NaN=0
idV	shape=(24, 50)	NaN=0
ieDA_m	shape=(24, 50)	NaN=0
ieDA_p	shape=(24, 50)	NaN=0
pIB_m	shape=(24, 50)	NaN=0
pIB_p	shape=(24, 50)	NaN=0
pPV	shape=(24, 50)	NaN=0
pW	shape=(24, 50)	NaN=0
rD	shape=(24, 50)	NaN=0
rD_B	shape=(24, 50)	NaN=0
rD_FD	shape=(24, 50)	NaN=0
rU	shape=(24, 50)	NaN=0
rU_B	shape=(24, 50)	NaN=0
rU_FD	shape=(24, 50)	NaN=0
socV	shape=(24, 50)	NaN=0
var_afd_m	shape=(24, 50)	NaN=0
var_afd_p	shape=(24, 50)	NaN=0
var_fd	shape=(24, 50)	NaN=0

2b ? How pIB_p and pIB_m are calculated

The model defines both variables as non-negative and enforces this balance in

model_builder.py :

```

pIB_p[t,q,s] - pIB_m[t,q,s] = eDA_m + pW + pPV + dV*dQ - eDA_p -
eIM - var_fd - cV*dQ

```

Therefore the saved values are the positive and negative parts of the same net imbalance:

- `pIB_p = max(balance, 0)`
- `pIB_m = max(-balance, 0)`

If hydrogen variables are loaded, the model also adds `+ eFC_on - eEL - eCOMP + eFC_sb` to the balance.

```
In [4]: try:
        display
    except NameError:
        display = print

    def _align_for_calc(data, names):
        dfs = [data[n] for n in names if n in data]
        if not dfs:
            return {}
        common_index = dfs[0].index
        common_cols = dfs[0].columns
        for df in dfs[1:]:
            common_index = common_index.intersection(df.index)
            common_cols = common_cols.intersection(df.columns)
        return {n: data[n].loc[common_index, common_cols] for n in names if n in data}

    needed = ["eDA_m", "pW", "pPV", "dV", "eDA_p", "eIM", "var_fd", "cV"]
    available = _align_for_calc(data, needed + ["eFC_on", "eEL", "eCOMP", "eFC_sb"])
    missing = [n for n in needed if n not in available]

    if missing:
        print("Cannot recalculate pIB_p/pIB_m. Missing variables:", missing)
    else:
        sample_df = available[needed[0]]
        if isinstance(sample_df.index, pd.MultiIndex) and "q" in sample_df.index.names:
            n_q = sample_df.index.get_level_values("q").nunique()
        else:
            n_q = max(1, sample_df.shape[0] // CONFIG["n_periods"])
        dQ = 1 / n_q

        balance = (
            available["eDA_m"]
            + available["pW"]
            + available["pPV"]
            + available["dV"] * dQ
            - available["eDA_p"]
            - available["eIM"]
            - available["var_fd"]
            - available["cV"] * dQ
        )
        if {"eFC_on", "eEL", "eCOMP", "eFC_sb"}.issubset(available):
            balance = balance + available["eFC_on"] - available["eEL"] - available["eCOMP"] - available["eFC_sb"]

        pIB_p_calc = balance.clip(lower=0)
        pIB_m_calc = (-balance).clip(lower=0)
```

```

print(f"Detected n_q={n_q}; using dQ={dQ:.4f}")
display(pd.DataFrame({
    "balance_mean": balance.mean(axis=1),
    "pIB_p_calc_mean": pIB_p_calc.mean(axis=1),
    "pIB_m_calc_mean": pIB_m_calc.mean(axis=1),
}).head(12))

if "pIB_p" in available and "pIB_m" in available:
    print("Absolute error vs stored pIB_p:")
    display((available["pIB_p"] - pIB_p_calc).abs().stack().describe())
    print("Absolute error vs stored pIB_m:")
    display((available["pIB_m"] - pIB_m_calc).abs().stack().describe())

```

Cannot recalculate pIB_p/pIB_m. Missing variables: ['dV', 'cV']

3 · Descriptive Statistics

Summary table: mean, std, min, P10, median, P90, max, % zeros and % negatives across all time steps and scenarios.

```

In [5]: rows = []
for var, df in sorted(data.items()):
    flat = df.values.flatten()
    rows.append({
        "Variable" : var,
        "Unit"      : UNITS.get(var, "-"),
        "T"         : df.shape[0],
        "S"         : df.shape[1],
        "Mean"      : np.nanmean(flat),
        "Std"       : np.nanstd(flat),
        "Min"       : np.nanmin(flat),
        "P10"       : np.nanpercentile(flat,10),
        "Median"    : np.nanmedian(flat),
        "P90"       : np.nanpercentile(flat,90),
        "Max"       : np.nanmax(flat),
        "% zero"    : 100*np.mean(flat==0),
        "% neg"     : 100*np.mean(flat<0),
    })

summary = pd.DataFrame(rows).set_index("Variable")
pd.set_option("display.float_format", "{:.4f}".format)
display(summary)

```

	Unit	T	S	Mean	Std	Min	P10	Median	P90	Max
Variable										
cV_dV	MW	24	50	0.0091	0.2161	-0.3930	-0.2835	0.0218	0.2820	0.4027
eDA_m	MWh	24	50	0.0882	0.0294	0.0115	0.0502	0.0873	0.1278	0.1729
eDA_p	MWh	24	50	0.8822	0.2904	0.1346	0.4999	0.8810	1.2591	1.6180
eIM	MWh	24	50	-0.0001	0.0350	-0.0679	-0.0472	0.0001	0.0464	0.0644
idV	-	24	50	0.4867	0.4998	0.0000	0.0000	0.0000	1.0000	1.0000
ieDA_m	MWh	24	50	0.0441	0.0488	0.0000	0.0000	0.0194	0.1199	0.1750
ieDA_p	MWh	24	50	0.0866	0.0975	0.0000	0.0000	0.0380	0.2410	0.3395
plB_m	MWh	24	50	0.0543	0.0591	0.0000	0.0000	0.0285	0.1497	0.2214
plB_p	MWh	24	50	0.0728	0.0800	0.0000	0.0000	0.0352	0.1968	0.2998
pPV	MW	24	50	0.6268	0.7522	0.0000	0.0000	0.1278	1.8733	2.1995
pW	MW	24	50	0.6035	0.3271	0.0000	0.1630	0.6371	1.0265	1.2415
rD	MW	24	50	0.2048	0.0793	0.0244	0.0994	0.2057	0.3088	0.3760
rD_B	MW	24	50	0.0657	0.0326	0.0000	0.0167	0.0709	0.1049	0.1229
rD_FD	MW	24	50	0.0647	0.0325	0.0000	0.0164	0.0721	0.1018	0.1320
rU	MW	24	50	0.2004	0.0766	0.0250	0.0986	0.1976	0.3031	0.3755
rU_B	MW	24	50	0.0622	0.0338	0.0000	0.0105	0.0682	0.1016	0.1248
rU_FD	MW	24	50	0.0643	0.0323	0.0000	0.0169	0.0689	0.1034	0.1377
socV	MWh	24	50	0.5011	0.2095	0.1373	0.2170	0.4980	0.7816	0.8665
var_afd_m	MW	24	50	0.0651	0.0330	0.0000	0.0146	0.0721	0.1042	0.1378
var_afd_p	MW	24	50	0.0609	0.0338	0.0000	0.0098	0.0660	0.1022	0.1420
var_fd	MW	24	50	0.9877	0.3222	0.2285	0.5651	0.9800	1.4106	1.7604

Per-timestep spread (mean over scenarios → variability over time)

```
In [6]: spread = []
for var, df in sorted(data.items()):
    spread.append({
        "Variable"      : var,
        "Unit"          : UNITS.get(var, "-"),
        "Time-mean of mean" : df.mean(axis=1).mean(),
        "Std of mean"      : df.mean(axis=1).std(),
        "Avg scen spread"  : df.std(axis=1).mean(),
        "Max scen spread"  : df.std(axis=1).max(),
    })
pd.DataFrame(spread).set_index("Variable")
```

Out[6]:

	Unit	Time-mean of mean	Std of mean	Avg scen spread	Max scen spread
Variable					
cV_dV	MW	0.0091	0.2168	0.0406	0.0490
eDA_m	MWh	0.0882	0.0261	0.0147	0.0171
eDA_p	MWh	0.8822	0.2630	0.1348	0.1793
eIM	MWh	-0.0001	0.0353	0.0058	0.0071
idV	-	0.4867	0.0838	0.4980	0.5051
ieDA_m	MWh	0.0441	0.0482	0.0120	0.0174
ieDA_p	MWh	0.0866	0.0971	0.0211	0.0326
plB_m	MWh	0.0543	0.0572	0.0182	0.0265
plB_p	MWh	0.0728	0.0780	0.0230	0.0337
pPV	MW	0.6268	0.7645	0.0726	0.1058
pW	MW	0.6035	0.3201	0.0944	0.1100
rD	MW	0.2048	0.0740	0.0324	0.0401
rD_B	MW	0.0657	0.0314	0.0109	0.0137
rD_FD	MW	0.0647	0.0313	0.0107	0.0133
rU	MW	0.2004	0.0717	0.0310	0.0355
rU_B	MW	0.0622	0.0330	0.0101	0.0124
rU_FD	MW	0.0643	0.0309	0.0113	0.0130
socV	MWh	0.5011	0.2125	0.0251	0.0284
var_afd_m	MW	0.0651	0.0317	0.0113	0.0134
var_afd_p	MW	0.0609	0.0330	0.0100	0.0125
var_fd	MW	0.9877	0.2923	0.1486	0.1824

4 · Fig 1 – Renewable Generation & BESS SoC

What to look for:

- Width of the CI band → how uncertain is each generation source.
- SoC dip/peak alignment with PV peak → confirms solar-driven charging.
- If SoC CI is wide, the BESS dispatch varies significantly across scenarios.

```
In [7]: fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(12,7), sharex=True)
fig.suptitle("Fig 1 – Renewable Generation & BESS SoC", fontweight="bold")
t = _t(next(iter(data.values())))

if "pPV" in data: _band(ax1, t, data["pPV"], COLORS["pv"], "PV (pPV)")
```

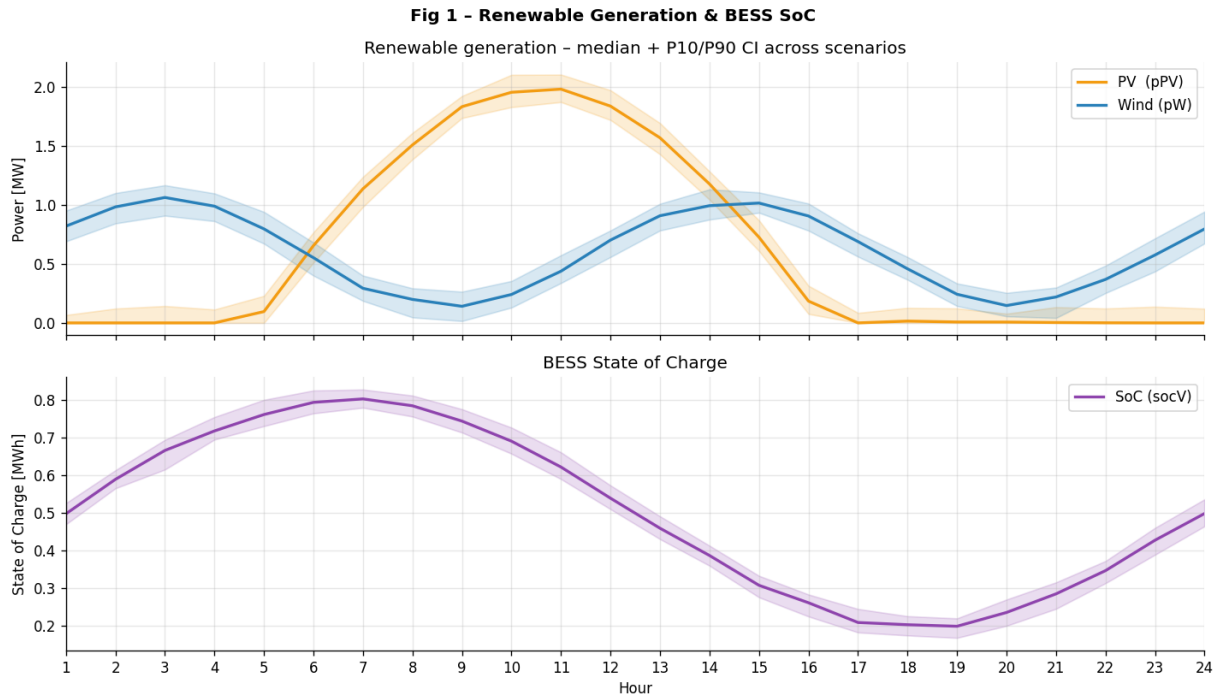
```

if "pW" in data: _band(ax1, t, data["pW"], COLORS["wind"], "Wind (pW)")
ax1.set_ylabel("Power [MW]"); ax1.legend(); ax1.grid(alpha=0.3)
ax1.set_title("Renewable generation – median + P10/P90 CI across scenarios")

if "socV" in data:
    _band(ax2, t, data["socV"], COLORS["bess"], "SoC (socV)")
    _annotate_socv_initial(ax2)
ax2.set_ylabel("State of Charge [MWh]"); ax2.legend(); ax2.grid(alpha=0.3)
ax2.set_title("BESS State of Charge")
ax2.set_xlabel(CONFIG["period_label"])
_format_hour_axis(ax2, t if "t" in globals() else None)

plt.tight_layout(); _save(fig, "fig1_generation"); plt.show()

```



5 · Fig 2 – BESS Operation (cV_dV & idV)

What to look for:

- $cV_dV > 0$ → BESS is charging; < 0 → discharging. Diverging colours make this instant to read.
- idV bar height → in what fraction of scenarios is the BESS actually dispatched to discharge at that hour.
- Discrepancy between median $cV_dV < 0$ and low idV → some scenarios charge while others discharge (high uncertainty in dispatch decision).

```

In [8]: fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(12,7), sharex=True)
fig.suptitle("Fig 2 – BESS Operation", fontweight="bold")
t = _t(next(iter(data.values())))

if "cV_dV" in data:
    med, lo, hi = _ci(data["cV_dV"])

```

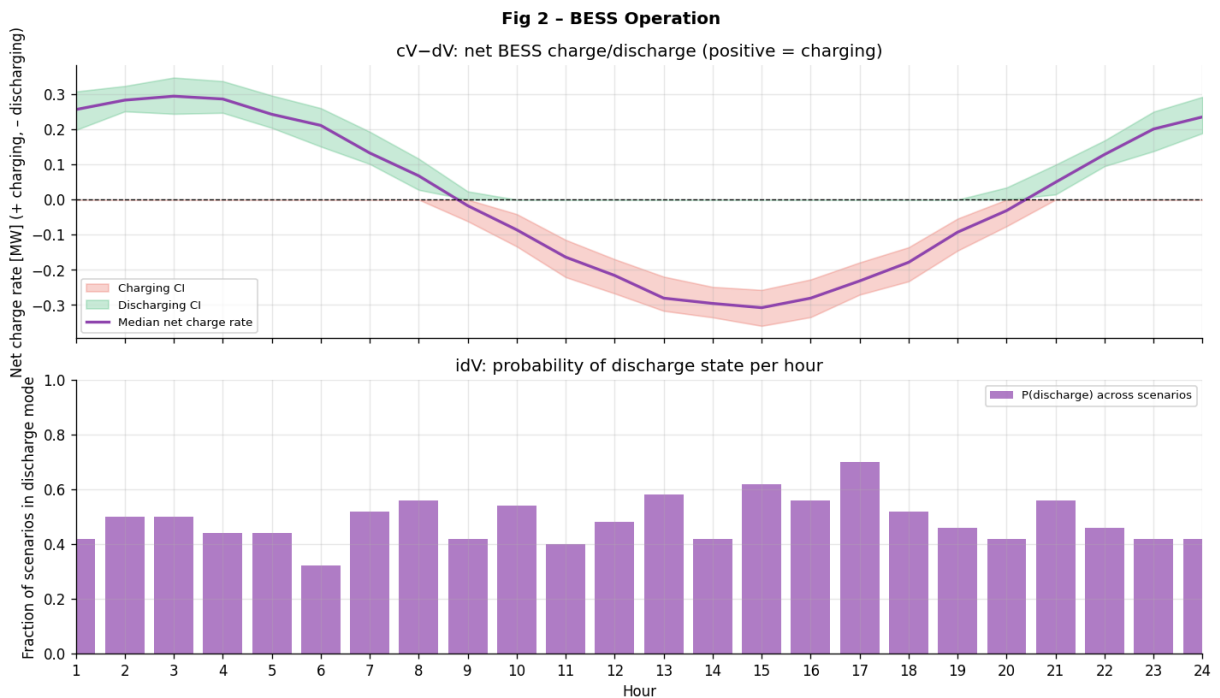
```

ax1.fill_between(t, np.minimum(lo,0), np.minimum(hi,0), color=COLORS["ne
ax1.fill_between(t, np.maximum(lo,0), np.maximum(hi,0), color=COLORS["pc
ax1.plot(t, med, color=COLORS["bess"], lw=2, label="Median net charge ra
ax1.axhline(0, color="k", lw=0.8, ls="--")
ax1.set_ylabel("Net charge rate [MW] (+ charging, - discharging)")
ax1.legend(fontsize=8); ax1.grid(alpha=0.3)
ax1.set_title("cV-dV: net BESS charge/discharge (positive = charging)")

if "idV" in data:
    ax2.bar(t, data["idV"].mean(axis=1), color=COLORS["bess"], alpha=0.7, la
    ax2.set_ylim(0,1)
    ax2.set_ylabel("Fraction of scenarios in discharge mode")
    ax2.legend(fontsize=8); ax2.grid(alpha=0.3)
    ax2.set_title("idV: probability of discharge state per hour")
    ax2.set_xlabel(CONFIG["period_label"])
    _format_hour_axis(ax2, t if "t" in globals() else None)

plt.tight_layout(); _save(fig, "fig2_bess_operation"); plt.show()

```



6 · Fig 3 – Flexible Demand

What to look for:

- Panel (a): total `var_fd` — does the EC smooth its demand effectively?
- Panel (b): mirrored `var_afd_p` / `var_afd_m` — are both shifts active simultaneously (arbitrage) or mutually exclusive?
- Panel (c): boxplot of **net shift** — wide boxes indicate high scenario uncertainty in how the FD is used. Outliers suggest extreme activation events.

```

In [9]: fig, (ax1, ax2, ax3) = plt.subplots(3, 1, figsize=(12,10), sharex=True)
fig.suptitle("Fig 3 – Flexible Demand", fontweight="bold")
t = _t(next(iter(data.values())))

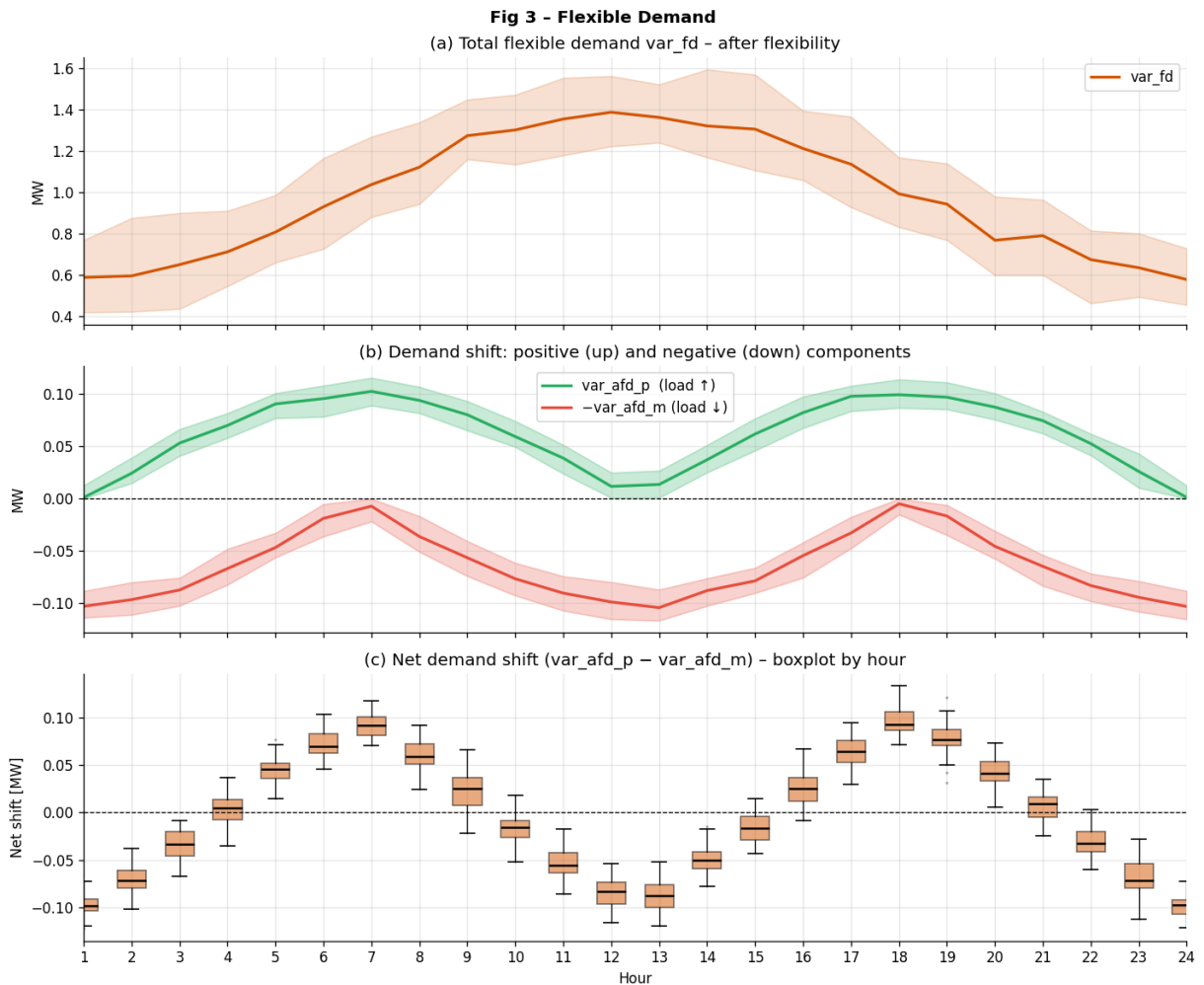
# (a) total flexible demand
if "var_fd" in data:
    _band(ax1, t, data["var_fd"], COLORS["fd"], "var_fd")
ax1.set_ylabel("MW"); ax1.legend(); ax1.grid(alpha=0.3)
ax1.set_title("(a) Total flexible demand var_fd – after flexibility")

# (b) positive / negative shifts mirrored
if "var_afd_p" in data and "var_afd_m" in data:
    med_p, lp, hp = _ci(data["var_afd_p"])
    med_m, lm, hm = _ci(data["var_afd_m"])
    ax2.fill_between(t, lp, hp, color=COLORS["pos"], alpha=0.25)
    ax2.plot(t, med_p, color=COLORS["pos"], lw=2, label="var_afd_p (load ↑)")
    ax2.fill_between(t, -hm, -lm, color=COLORS["neg"], alpha=0.25)
    ax2.plot(t, -med_m, color=COLORS["neg"], lw=2, label="-var_afd_m (load ↓)")
    ax2.axhline(0, color="k", lw=0.8, ls="--")
ax2.set_ylabel("MW"); ax2.legend(); ax2.grid(alpha=0.3)
ax2.set_title("(b) Demand shift: positive (up) and negative (down) component")

# (c) net shift boxplot per hour
if "var_afd_p" in data and "var_afd_m" in data:
    net = (data["var_afd_p"] - data["var_afd_m"]).values
    ax3.boxplot([net[i,:] for i in range(len(t))],
                positions=t, widths=0.6, patch_artist=True,
                boxprops=dict(facecolor=COLORS["fd"], alpha=0.5),
                medianprops=dict(color="k", lw=1.5),
                whiskerprops=dict(lw=0.8),
                flierprops=dict(marker=".", ms=2, alpha=0.3))
    ax3.axhline(0, color="k", lw=0.8, ls="--")
ax3.set_ylabel("Net shift [MW]"); ax3.grid(alpha=0.3)
ax3.set_title("(c) Net demand shift (var_afd_p – var_afd_m) – boxplot by hour")
ax3.set_xlabel(CONFIG["period_label"])
_format_hour_axis(ax3, t if "t" in globals() else None)

plt.tight_layout(); _save(fig, "fig3_flexible_demand"); plt.show()

```



7 · Fig 4 – Day-Ahead Commitment

What to look for:

- **Panel (a):** committed energy in DA
- **Panel (b):** committed energy in RM
- **Panel (c):** committed energy in IM1
- **Panel (d):** committed energy in IM2
- **Panel (e):** committed energy in IM3

```
In [10]: fig, axs = plt.subplots(5, 1, figsize=(12,16), sharex=True)
fig.suptitle("Fig 4 – Day-Ahead Commitment by Stage △ KEY FIGURE",
             fontweight="bold", color=COLORS["neg"])
t = _t(next(iter(data.values())))

# (a) committed energy in DA
if "eDA_p" in data:
    med, lo, hi = _ci(data["eDA_p"])
    axs[0].fill_between(t, lo, hi, color=COLORS["da"], alpha=0.2)
    axs[0].plot(t, med, lw=2, color=COLORS["da"], label="eDA_p")
if "eDA_m" in data:
    med, lo, hi = _ci(data["eDA_m"])
```

```

    axs[0].fill_between(t, lo, hi, color=COLORS["neg"], alpha=0.2)
    axs[0].plot(t, med, lw=2, color=COLORS["neg"], label="eDA_m")
axs[0].set_ylabel("MWh"); axs[0].legend(fontsize=8); axs[0].grid(alpha=0.3)
axs[0].set_title("(a) Committed energy in DA")

# (b) committed energy in RM
if "rU" in data:
    med, lo, hi = _ci(data["rU"])
    axs[1].fill_between(t, lo, hi, color=COLORS["reserve"], alpha=0.2)
    axs[1].plot(t, med, lw=2, color=COLORS["reserve"], label="rU")
if "rD" in data:
    med, lo, hi = _ci(data["rD"])
    axs[1].fill_between(t, lo, hi, color=COLORS["neg"], alpha=0.2)
    axs[1].plot(t, med, lw=2, color=COLORS["neg"], label="rD")
axs[1].set_ylabel("MW"); axs[1].legend(fontsize=8); axs[1].grid(alpha=0.3)
axs[1].set_title("(b) Committed energy in RM")

# (c) committed energy in IM1
if "eIM_1" in data:
    med, lo, hi = _ci(data["eIM_1"])
    axs[2].fill_between(t, lo, hi, color=COLORS["pos"], alpha=0.2)
    axs[2].plot(t, med, lw=2, color=COLORS["pos"], label="eIM_1")
axs[2].set_ylabel("MWh"); axs[2].legend(fontsize=8); axs[2].grid(alpha=0.3)
axs[2].set_title("(c) Committed energy in IM1")

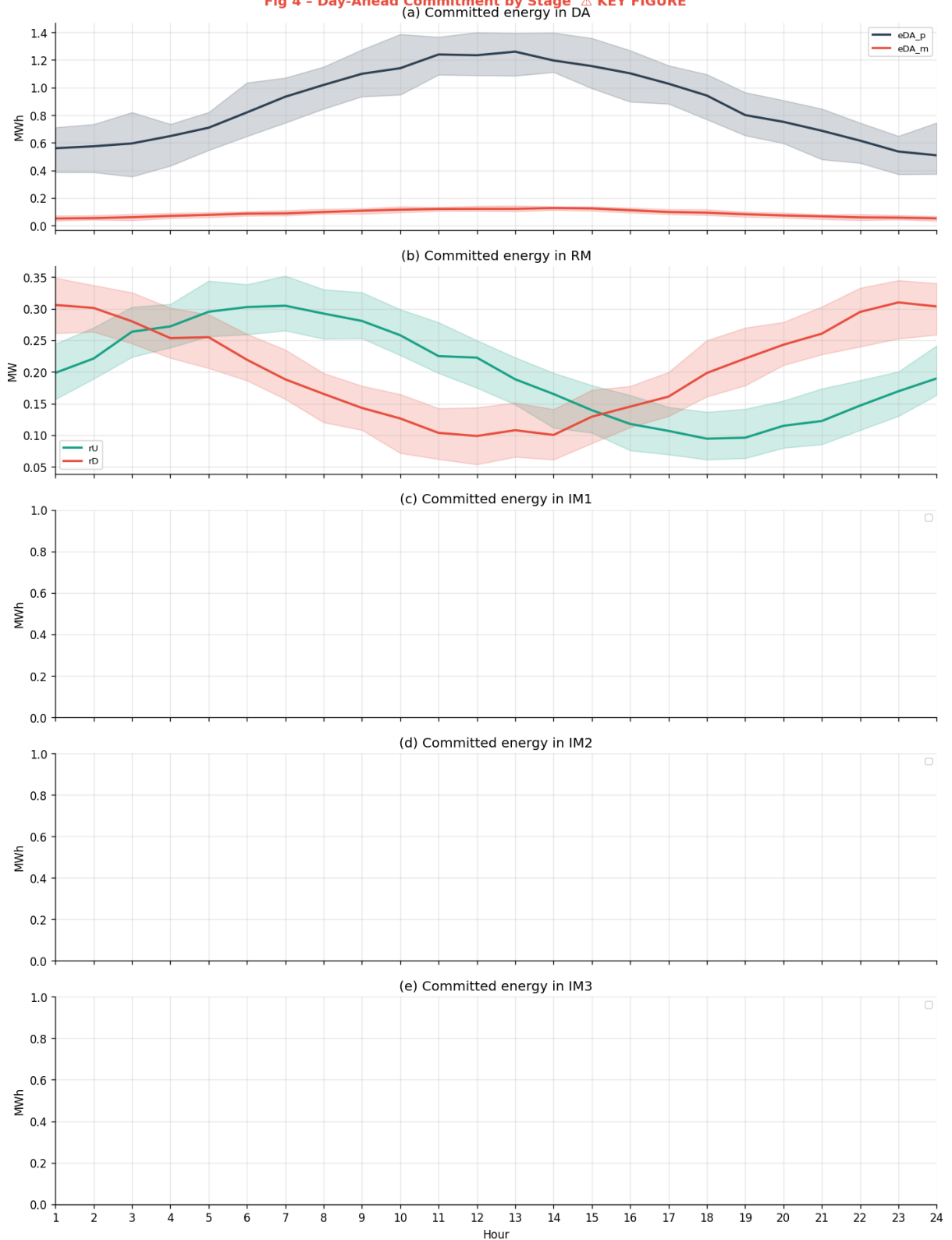
# (d) committed energy in IM2
if "eIM_2" in data:
    med, lo, hi = _ci(data["eIM_2"])
    axs[3].fill_between(t, lo, hi, color=COLORS["wind"], alpha=0.2)
    axs[3].plot(t, med, lw=2, color=COLORS["wind"], label="eIM_2")
axs[3].set_ylabel("MWh"); axs[3].legend(fontsize=8); axs[3].grid(alpha=0.3)
axs[3].set_title("(d) Committed energy in IM2")

# (e) committed energy in IM3
if "eIM_3" in data:
    med, lo, hi = _ci(data["eIM_3"])
    axs[4].fill_between(t, lo, hi, color=COLORS["median"], alpha=0.2)
    axs[4].plot(t, med, lw=2, color=COLORS["median"], label="eIM_3")
axs[4].set_ylabel("MWh"); axs[4].legend(fontsize=8); axs[4].grid(alpha=0.3)
axs[4].set_title("(e) Committed energy in IM3")
axs[4].set_xlabel(CONFIG["period_label"])
_format_hour_axis(axs[4], t if "t" in globals() else None)

plt.tight_layout(); _save(fig, "fig4_da_commitment"); plt.show()

```

Fig 4 - Day-Ahead Commitment by Stage ▲ KEY FIGURE



8 · Fig 5 – Reserve Provision (Upward & Downward)

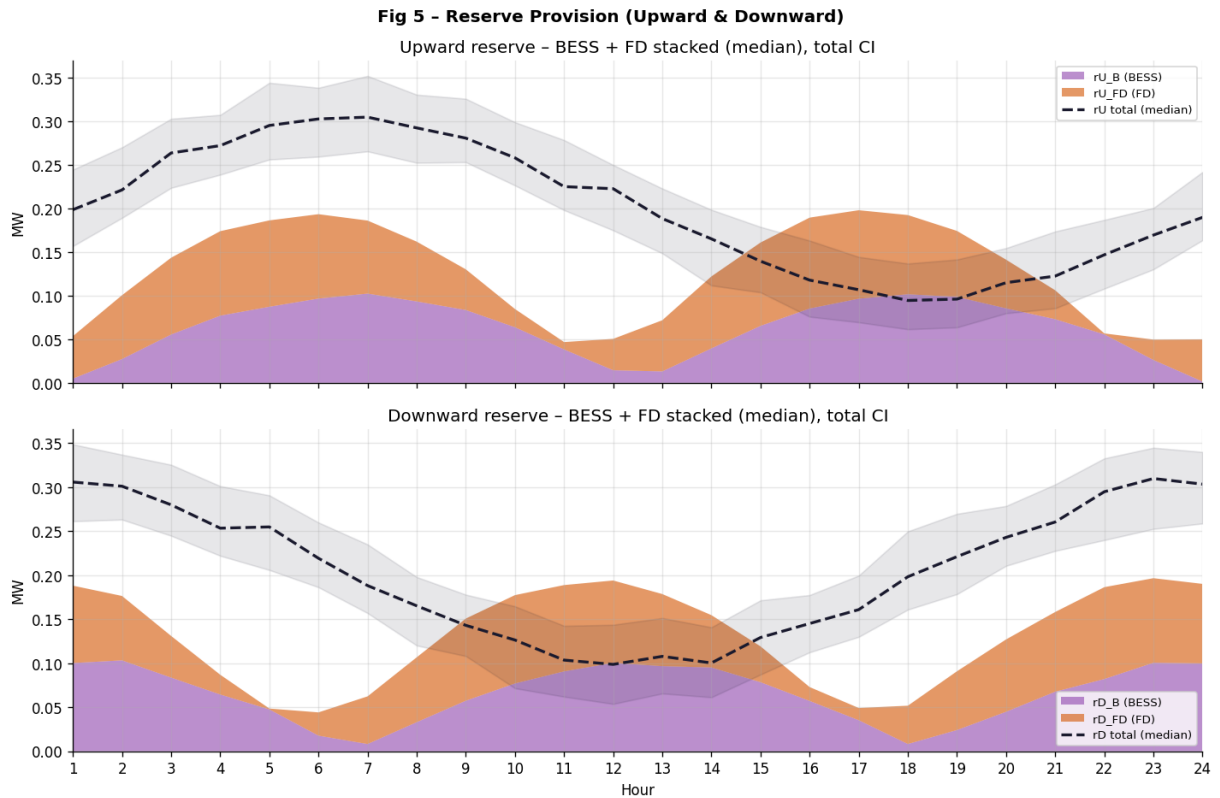
What to look for:

- Which asset (BESS vs FD) carries the bulk of reserve provision at each hour.
- Dashed total line vs. stacked area → if total > BESS + FD something is off (data check).
- Wide CI on the total reserve → significant scenario uncertainty in reserve matching.

```
In [11]: fig, (ax_u, ax_d) = plt.subplots(2, 1, figsize=(12,8), sharex=True)
fig.suptitle("Fig 5 – Reserve Provision (Upward & Downward)", fontweight="bold")
t = _t(next(iter(data.values()))))

for ax, label, r_tot, r_b, r_fd in [
    (ax_u, "Upward", "rU", "rU_B", "rU_FD"),
    (ax_d, "Downward", "rD", "rD_B", "rD_FD"),
]:
    med_b = data[r_b].median(axis=1).values if r_b in data else np.zeros(
    med_fd = data[r_fd].median(axis=1).values if r_fd in data else np.zeros(
    ax.stackplot(t, med_b, med_fd,
                labels=[f"{r_b} (BESS)", f"{r_fd} (FD)"],
                colors=[COLORS["bess"], COLORS["fd"]], alpha=0.6)
    if r_tot in data:
        med_tot, lo, hi = _ci(data[r_tot])
        ax.plot(t, med_tot, color=COLORS["median"], lw=2, ls="--",
                label=f"{r_tot} total (median)")
        ax.fill_between(t, lo, hi, color=COLORS["median"], alpha=0.1)
    ax.set_ylabel("MW"); ax.legend(fontsize=8); ax.grid(alpha=0.3)
    ax.set_title(f"{label} reserve – BESS + FD stacked (median), total CI")

ax_d.set_xlabel(CONFIG["period_label"])
_format_hour_axis(ax_d, t if "t" in globals() else None)
plt.tight_layout(); _save(fig, "fig5_reserves"); plt.show()
```



9 · Fig 6 – Imbalance & Intraday Position ⚠ KEY FIGURE

What to look for:

- **Panel (a):** mirrored pIB_p / pIB_m — does the EC mostly over-deliver or under-deliver?
- **Panel (b):** net imbalance bars — red hours = under-delivery (penalty exposure); green = over-delivery (spill or revenue).
- **Panel (c):** `eIM` — does intraday trading reduce imbalance at the same hours where panel (b) is red? If yes, the intraday market is effective.

```
In [12]: fig, (ax1, ax2, ax3) = plt.subplots(3, 1, figsize=(12,10), sharex=True)
fig.suptitle("Fig 6 – Imbalance & Intraday Position ⚠ KEY FIGURE", fontweight="bold")
t = _t(next(iter(data.values())))

# (a) pIB_p / pIB_m mirrored
if "pIB_p" in data:
    _band(ax1, t, data["pIB_p"], COLORS["pos"], "pIB_p (over-delivery)")
if "pIB_m" in data:
    med_m, lo_m, hi_m = _ci(data["pIB_m"])
    ax1.fill_between(t, -hi_m, -lo_m, color=COLORS["neg"], alpha=0.18)
    ax1.plot(t, -med_m, color=COLORS["neg"], lw=2, label="-pIB_m (under-delivery)")
ax1.axhline(0, color="k", lw=0.8, ls="--")
ax1.set_ylabel("MWh"); ax1.legend(fontsize=8); ax1.grid(alpha=0.3)
ax1.set_title("(a) Positive (over-delivery) and negative (under-delivery) imbalances")
```

```

# (b) net imbalance bar
if "pIB_p" in data and "pIB_m" in data:
    net_med = (data["pIB_p"] - data["pIB_m"]).median(axis=1).values
    ax2.bar(t, net_med, color=[COLORS["pos"] if v>=0 else COLORS["neg"] for
    ax2.axhline(0, color="k", lw=0.8)
ax2.set_ylabel("Net imbalance [MWh]"); ax2.grid(alpha=0.3)
ax2.set_title("(b) Net imbalance pIB_p - pIB_m - median across scenarios")

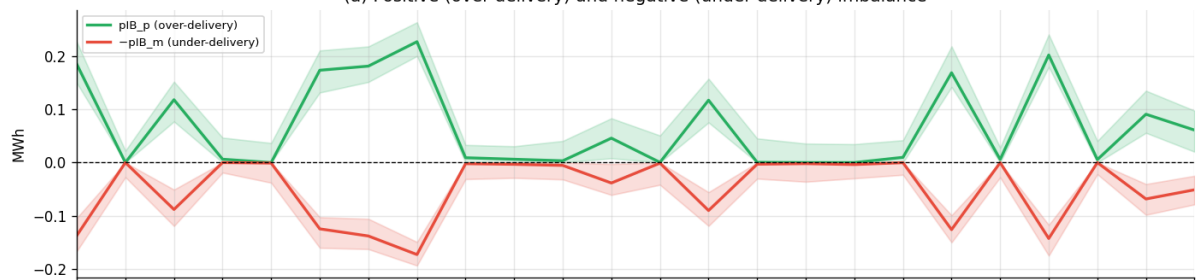
# (c) eIM intraday
if "eIM" in data:
    med_im, lo_im, hi_im = _ci(data["eIM"])
    ax3.fill_between(t, lo_im, hi_im, color=COLORS["wind"], alpha=0.25)
    ax3.plot(t, med_im, color=COLORS["wind"], lw=2, label="eIM (intraday)")
    ax3.axhline(0, color="k", lw=0.8, ls="--")
ax3.set_ylabel("MWh"); ax3.legend(fontsize=8); ax3.grid(alpha=0.3)
ax3.set_title("(c) Intraday energy eIM - positive = buy, negative = sell")
ax3.set_xlabel(CONFIG["period_label"])
_format_hour_axis(ax3, t if "t" in globals() else None)

plt.tight_layout(); _save(fig, "fig6_imbalance"); plt.show()

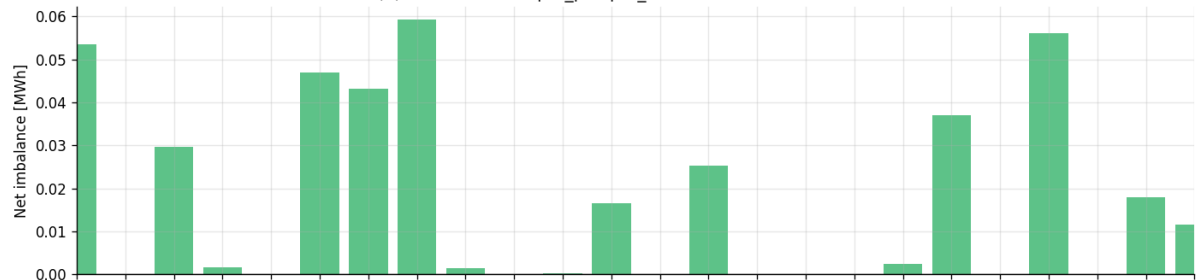
```

Fig 6 - Imbalance & Intraday Position △ **KEY FIGURE**

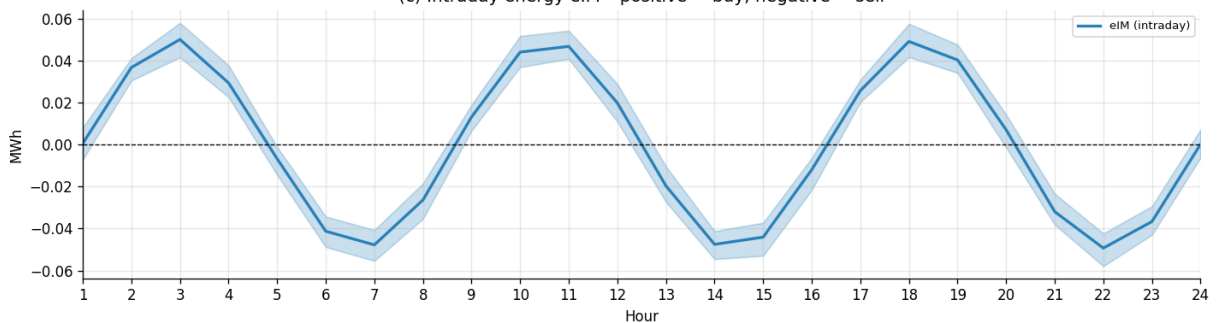
(a) Positive (over-delivery) and negative (under-delivery) imbalance



(b) Net imbalance pIB_p - pIB_m - median across scenarios



(c) Intraday energy eIM - positive = buy, negative = sell



10 · Fig 7 – Scenario Fan Charts (spaghetti)

What to look for:

- Are traces clustered (low spread) or spread widely (high uncertainty)?
- Bimodal shapes → two distinct operating regimes across scenarios.
- Extreme outlier traces → which scenario drives the worst case?

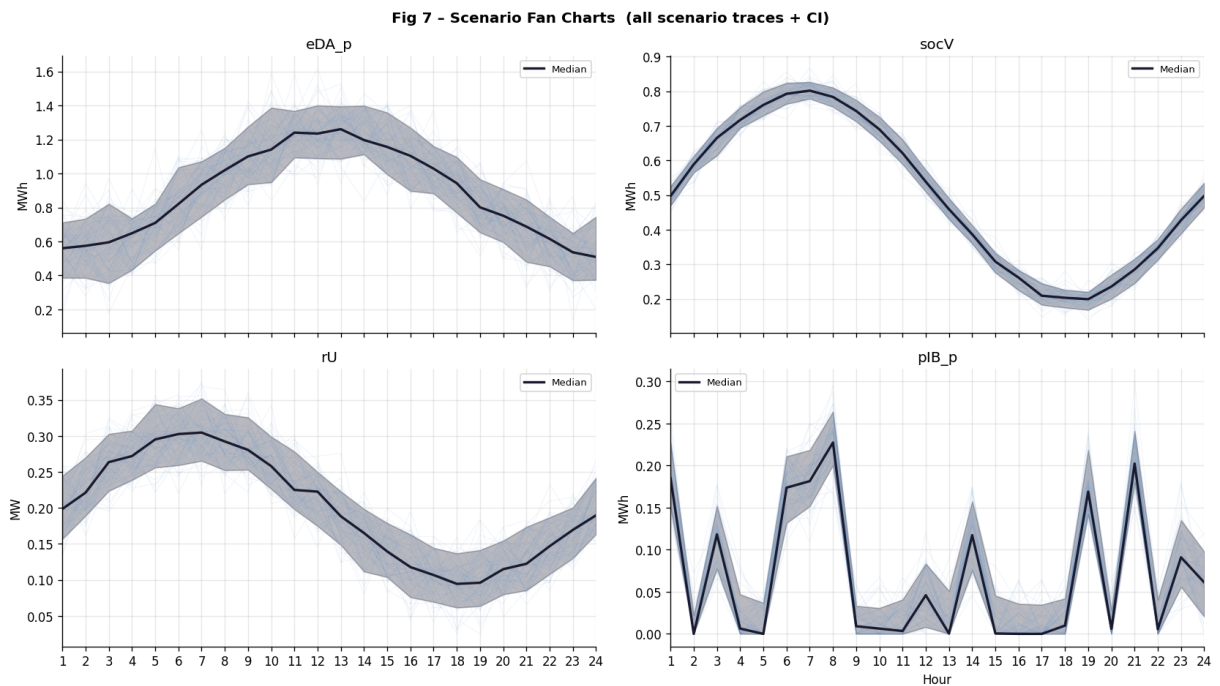
You can change `vars_to_show` to inspect any subset of variables.

```
In [13]: vars_to_show = ["eDA_p", "socV", "rU", "pIB_p"] # ← edit here
vars_to_show = [v for v in vars_to_show if v in data]

ncols = 2
nrows = int(np.ceil(len(vars_to_show)/ncols))
fig, axes = plt.subplots(nrows, ncols, figsize=(14, 4*nrows), sharex=True)
axes = np.array(axes).flatten()
fig.suptitle("Fig 7 - Scenario Fan Charts (all scenario traces + CI)", fontt = _t(next(iter(data.values()))))

for ax, var in zip(axes, vars_to_show):
    df = data[var]
    for col in df.columns:
        ax.plot(t, df[col].values, color=COLORS["ci_fill"], lw=0.4, alpha=0.
        _band(ax, t, df, COLORS["median"], "Median", alpha=0.3)
    if var == "socV":
        _annotate_socv_initial(ax)
    ax.set_title(var); ax.set_ylabel(UNITS.get(var, "-")); ax.grid(alpha=0.3)

for ax in axes[len(vars_to_show):]: ax.set_visible(False)
axes[-1].set_xlabel(CONFIG["period_label"])
_format_hour_axis(axes[-1], t if "t" in globals() else None)
plt.tight_layout(); _save(fig, "fig7_scenario_fans"); plt.show()
```



11 ? Market Benefit Distribution

Boxplot of total scenario benefit by market. CI markers show P10-P90 across scenarios.

```
In [14]: def _aligned_product_total(*dfs):
    dfs = [df for df in dfs if df is not None]
    if not dfs:
        return None
    common_index = dfs[0].index
    common_cols = dfs[0].columns
    for df in dfs[1:]:
        common_index = common_index.intersection(df.index)
        common_cols = common_cols.intersection(df.columns)
    if len(common_index) == 0 or len(common_cols) == 0:
        return None
    prod = dfs[0].loc[common_index, common_cols].copy()
    for df in dfs[1:]:
        prod = prod * df.loc[common_index, common_cols]
    return prod.sum(axis=0)

market_benefits = {}

if {"LD", "eDA_p", "eDA_m"}.issubset(data):
    da_net = data["eDA_p"] - data["eDA_m"]
    market_benefits["DA"] = _aligned_product_total(data["LD"], da_net)

if {"LR", "rU", "rD"}.issubset(data):
    rm_reserve = data["rU"] + data["rD"]
    market_benefits["RM"] = _aligned_product_total(data["LR"], rm_reserve)

for i in [1, 2, 3]:
    li = f"LI_{i}"
    ei = f"eIM_{i}"
    if {li, ei}.issubset(data):
        market_benefits[f"IM{i}"] = _aligned_product_total(data[li], data[ei])

if {"lPIB", "pIB_p", "lNIB", "pIB_m"}.issubset(data):
    ib_income = _aligned_product_total(data["lPIB"], data["pIB_p"])
    ib_costs = _aligned_product_total(data["lNIB"], data["pIB_m"])
    if ib_income is not None and ib_costs is not None:
        common_cols = ib_income.index.intersection(ib_costs.index)
        market_benefits["IB"] = ib_income.loc[common_cols] - ib_costs.loc[common_cols]

market_order = ["DA", "RM", "IM1", "IM2", "IM3", "IB"]
market_benefits = {k: market_benefits[k].dropna() for k in market_order if k in market_benefits}

if not market_benefits:
    print("No market benefit data available for this run.")
else:
    labels = list(market_benefits.keys())
    values = [market_benefits[k].values for k in labels]
    positions = np.arange(1, len(labels) + 1)

    fig, ax = plt.subplots(figsize=(11, 5.5))
    bp = ax.boxplot(values, positions=positions, labels=labels, patch_artist=True)
    for patch, color in zip(bp["boxes"], [COLORS["da"], COLORS["reserve"], COLORS["im1"], COLORS["im2"], COLORS["im3"], COLORS["ib"]]):
```

```

        patch.set_facecolor(color)
        patch.set_alpha(0.25)
    for med in bp["medians"]:
        med.set_color(COLORS["median"])
        med.set_linewidth(2)

    ci_low, ci_high = CONFIG["ci_low"], CONFIG["ci_high"]
    medians = np.array([np.median(v) for v in values])
    lo = np.array([np.percentile(v, ci_low) for v in values])
    hi = np.array([np.percentile(v, ci_high) for v in values])
    ax.errorbar(positions, medians, yerr=[medians - lo, hi - medians], fmt="
        color=COLORS["median"], ecolor=COLORS["median"], capsize=5,
        label=f"Median with P{ci_low}-P{ci_high} CI")
    ax.axhline(0, color="#666666", lw=1, ls="--", alpha=0.7)
    ax.set_ylabel("Total scenario benefit [EUR]")
    ax.set_title("Market benefit distribution by scenario", fontweight="bold")
    ax.grid(axis="y", alpha=0.3)
    ax.legend(fontsize=8)
    plt.tight_layout(); _save(fig, "fig_market_benefits_boxplot"); plt.show()

benefit_summary = pd.DataFrame({
    "median": medians,
    f"P{ci_low}": lo,
    f"P{ci_high}": hi,
    "mean": [np.mean(v) for v in values],
}, index=labels)
display(benefit_summary)

```

No market benefit data available for this run.

11 · Fig 8 – Correlation Heatmap

What to look for:

- Strong positive correlation (dark blue) between PV/Wind and reserve/FD → renewable surplus drives reserve provision.
- Strong negative correlation between socV and pIB_m → better SoC management reduces imbalance penalties.
- Any unexpected high correlation that reveals hidden structural coupling in the model.

```

In [15]: op_data = _operational_ts_data(data)
med_df = pd.DataFrame({v: df.median(axis=1) for v, df in op_data.items()})
empty_vars = med_df.columns[med_df.isna().all()].tolist()
if empty_vars:
    print("Dropped empty variables from correlation heatmap:", empty_vars)
med_df = med_df.dropna(axis=1, how="all")
std = med_df.std(axis=0)
constant_vars = std[std <= 1e-12].index.tolist()
if constant_vars:
    print("Excluded constant variables from correlation heatmap:", constant_vars)
med_df = med_df.loc[:, std > 1e-12]
corr = med_df.corr(min_periods=2)

```

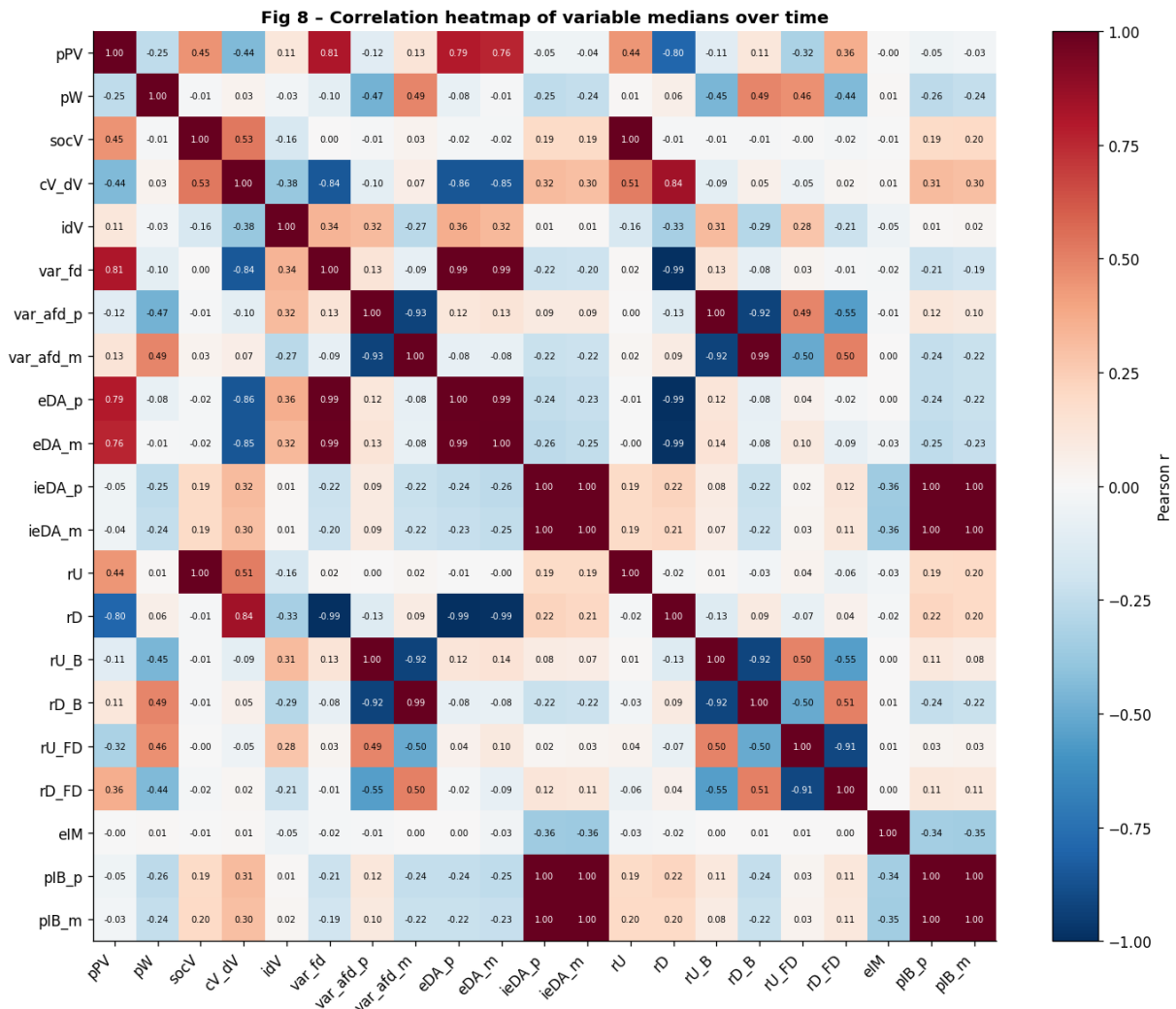
```

if corr.isna().any().any():
    print("Warning: correlation contains NaN values due to insufficient data")
print(f"Correlation heatmap uses {med_df.shape[1]} operational variables with

fig, ax = plt.subplots(figsize=(12,10))
cmap = plt.get_cmap("RdBu_r").copy()
cmap.set_bad("#d3d3d3")
im = ax.imshow(corr.values, cmap=cmap, vmin=-1, vmax=1, aspect="auto")
plt.colorbar(im, ax=ax, label="Pearson r")
ax.set_xticks(range(len(corr))); ax.set_xticklabels(corr.columns, rotation=45)
ax.set_yticks(range(len(corr))); ax.set_yticklabels(corr.columns)
for i in range(len(corr)):
    for j in range(len(corr)):
        val = corr.values[i,j]
        text = f"{val:.2f}" if not np.isnan(val) else "NaN"
        color = "white" if not np.isnan(val) and abs(val) > 0.6 else "black"
        ax.text(j, i, text, ha="center", va="center",
                fontsize=6, color=color)
ax.set_title("Fig 8 - Correlation heatmap of variable medians over time", fontweight="bold")
plt.tight_layout(); _save(fig, "fig8_correlation"); plt.show()

```

Correlation heatmap uses 21 operational variables with shared index 24 time points.



12 · Fig 9 – Commitment Failure Dashboard ⚠ KEY FIGURE

One-page answer to "when and how severely did the EC miss its day-ahead commitments?"

Sub-panel	What it shows
(a) Risk strip	colour = fraction of scenarios with <code>pIB_m > 0</code> ; red hours = most risky
(b) Shortfall bars	median magnitude of <code>pIB_m</code> (CI band)
(c) Delivered power	actual delivered power from <code>pPV + pW - cV_dV</code> when available; otherwise show <code>pIB_m</code> under-delivery proxy
(d) CDF	cumulative distribution of total daily under-delivery across all scenarios; P90 marks the tail risk

```
In [16]: fig = plt.figure(figsize=(14,12))
fig.suptitle("Fig 9 – Commitment Failure Dashboard\n"
            "(When and how severely did the EC miss its day-ahead commitment\n"
            fontweight="bold", color=COLORS["neg"], fontsize=13)

gs = gridspec.GridSpec(4, 2, width_ratios=[20, 1], wspace=0.08, hspace=0.5)
ax1 = fig.add_subplot(gs[0, 0])
cax = fig.add_subplot(gs[0, 1])
ax2 = fig.add_subplot(gs[1, 0], sharex=ax1)
ax3 = fig.add_subplot(gs[2, 0], sharex=ax1)
ax4 = fig.add_subplot(gs[3, 0])

t_arr = _t(next(iter(data.values())))
T = len(t_arr)
step = float(np.min(np.diff(t_arr))) if len(t_arr) > 1 else 1.0
x_edges = np.concatenate([t_arr - step/2, [t_arr[-1] + step/2]])
bar_width = min(0.8, step * 0.9)

# (a) risk heat-strip
prob = np.zeros(T)
if "pIB_m" in data:
    prob = (data["pIB_m"] > 1e-6).mean(axis=1).values
heat = ax1.pcolormesh(x_edges, [0, 1], prob[None, :], cmap="YlOrRd",
                     vmin=0, vmax=1, shading="auto")
fig.colorbar(heat, cax=cax, orientation="vertical")
cax.set_ylabel("P(shortfall)")
cax.yaxis.set_label_position("right")
cax.tick_params(axis="both", which="both", length=0)
ax1.set_yticks([])
ax1.set_title("(a) Risk strip: P(shortfall) per hour", fontsize=10)
ax1.set_xticks(np.arange(1, int(CONFIG["n_periods"]) + 1))
ax1.set_xticklabels([str(i) for i in range(1, int(CONFIG["n_periods"]) + 1)])
ax1.set_xlim(x_edges[0], x_edges[-1])
```

```

# (b) shortfall bars
if "pIB_m" in data:
    med, lo, hi = _ci(data["pIB_m"])
    ax2.fill_between(t_arr, lo, hi, color=COLORS["neg"], alpha=0.3)
    ax2.bar(t_arr, med, color=COLORS["neg"], alpha=0.7, width=bar_width)
ax2.set_ylabel("Shortfall [MWh]"); ax2.grid(alpha=0.3)
ax2.set_title("(b) Median under-delivery pIB_m (CI band)", fontsize=10)

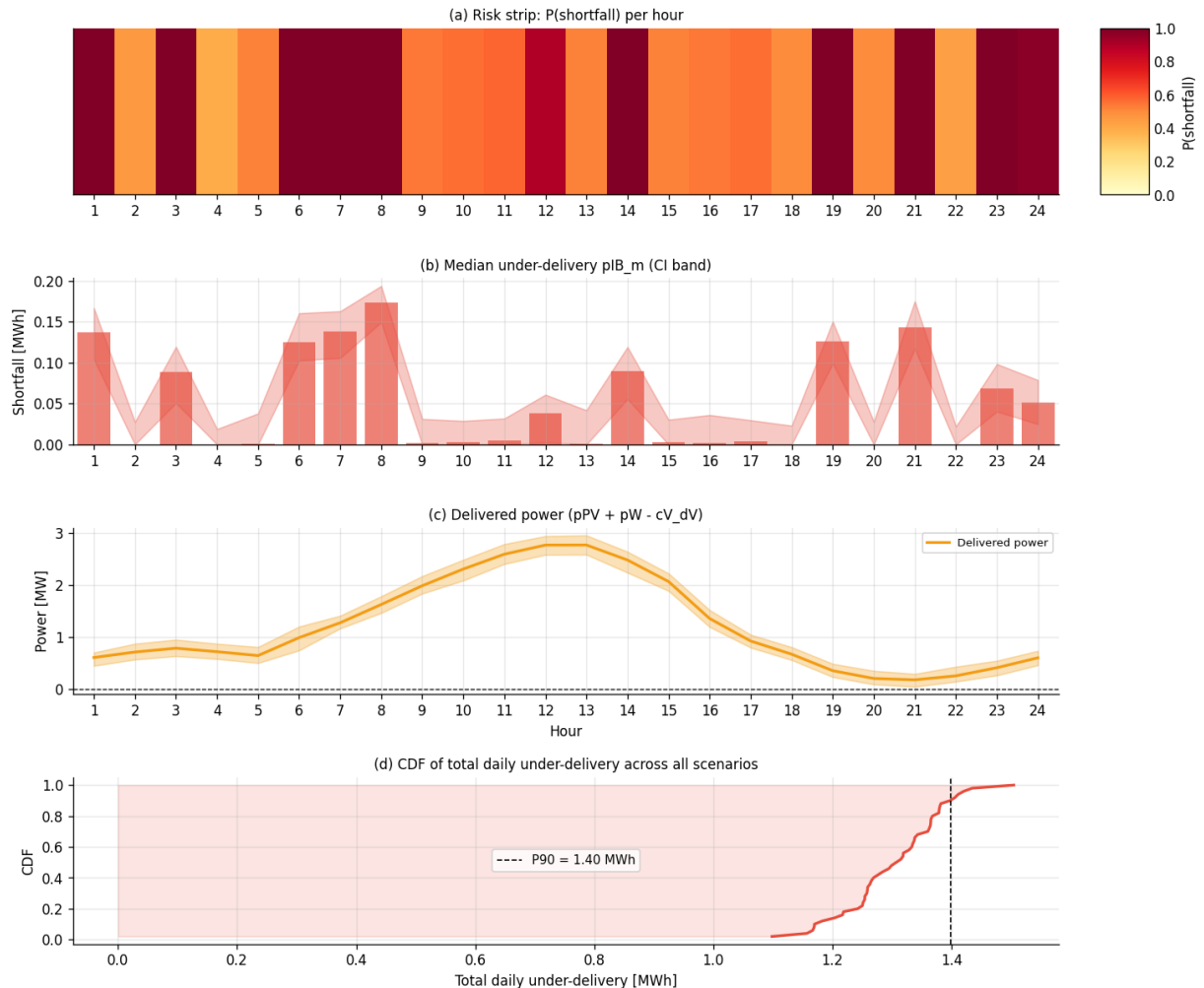
# (c) delivered power
if {"pPV", "pW", "cV_dV"}.issubset(data):
    delivered = data["pPV"] + data["pW"] - data["cV_dV"]
    med_del, lo_del, hi_del = _ci(delivered)
    ax3.fill_between(t_arr, lo_del, hi_del, color=COLORS["pv"], alpha=0.3)
    ax3.plot(t_arr, med_del, color=COLORS["pv"], lw=2, label="Delivered power")
    ax3.axhline(0, color="k", lw=0.8, ls="--")
    ax3.set_ylabel("Power [MW]"); ax3.set_xlabel(CONFIG["period_label"])
    ax3.set_xticks(np.arange(1, int(CONFIG["n_periods"]) + 1))
    ax3.set_xticklabels([str(i) for i in range(1, int(CONFIG["n_periods"]) + 1)])
    ax3.set_xlim(x_edges[0], x_edges[-1])
    ax3.legend(fontsize=8); ax3.grid(alpha=0.3)
    ax3.set_title("(c) Delivered power (pPV + pW - cV_dV)", fontsize=10)
else:
    if "pIB_m" in data:
        med_ib, lo_ib, hi_ib = _ci(data["pIB_m"])
        ax3.fill_between(t_arr, lo_ib, hi_ib, color=COLORS["imbal"], alpha=0.3)
        ax3.plot(t_arr, med_ib, color=COLORS["imbal"], lw=2, label="pIB_m median")
        ax3.set_ylabel("pIB_m [MWh]"); ax3.set_xlabel(CONFIG["period_label"])
        ax3.set_xticks(np.arange(1, int(CONFIG["n_periods"]) + 1))
        ax3.set_xticklabels([str(i) for i in range(1, int(CONFIG["n_periods"]) + 1)])
        ax3.set_xlim(x_edges[0], x_edges[-1])
        ax3.legend(fontsize=8); ax3.grid(alpha=0.3)
        ax3.set_title("(c) Negative imbalance pIB_m - proxy for under-delivery cost")

# (d) CDF of total daily under-delivery
if "pIB_m" in data:
    sf_tot = data["pIB_m"].sum(axis=0).values
    sf_s = np.sort(sf_tot)
    cdf = np.arange(1, len(sf_s)+1)/len(sf_s)
    ax4.plot(sf_s, cdf, color=COLORS["neg"], lw=2)
    ax4.fill_betweenx(cdf, 0, sf_s, color=COLORS["neg"], alpha=0.15)
    p90 = np.percentile(sf_s, 90)
    ax4.axvline(p90, color="k", ls="--", lw=1, label=f"P90 = {p90:.2f} MWh")
    ax4.set_xlabel("Total daily under-delivery [MWh]")
    ax4.set_ylabel("CDF"); ax4.legend(fontsize=9); ax4.grid(alpha=0.3)
    ax4.set_title("(d) CDF of total daily under-delivery across all scenarios")
else:
    ax4.text(0.5, 0.5, "No pIB_m data available for daily under-delivery CDF",
            ha="center", va="center", transform=ax4.transAxes)
    ax4.set_axis_off()

_save(fig, "fig9_commitment_dashboard"); plt.show()

```

Fig 9 - Commitment Failure Dashboard
(When and how severely did the EC miss its day-ahead commitments?)



13 · Export Descriptive Statistics to CSV

```
In [17]: if CONFIG["save_figs"]:
os.makedirs(CONFIG["fig_dir"], exist_ok=True)
out = os.path.join(CONFIG["fig_dir"], "descriptive_stats.csv")
summary.to_csv(out)
print(f"Saved → {out}")
else:
print("Set CONFIG['save_figs'] = True to export.")
```

Set CONFIG['save_figs'] = True to export.

14 · Batch Analysis — Scenario-Reduction Method Comparison (Same-Ren vs Different-Ren)

Everything above analyzes **one** simulation run in detail (per-scenario, per-timestep). This section instead compares **many** runs produced by

`codes/batch_run_scenarios.py` , which solves the DA market once per scenario-count folder for two scenario-reduction methodologies:

- **same_ren** - renewable profiles shared across the scenario tree
(`results_batch_same_ren.csv`)
- **dif_ren** - renewable profiles differ per scenario
(`results_batch_dif_ren.csv`)

Data sources:

1. The two batch CSVs (one row per solved scenario-count folder) — timing, resources, objective-function breakdown, and the `results_dir` of each run.
2. The per-run result files saved under `results/<scenario_folder>/DA/...` for that run (used here to compare Day-Ahead commitments, `eDA_p` / `eDA_m`).

This section is safe to run before all batch jobs finish: missing CSVs/rows/result files are skipped with a printed warning rather than raising an error, so re-running it later (once more scenario counts have completed) will simply show more data points.

```
In [18]: try:
          display
        except NameError:
          display = print

BATCH_METHOD_LABELS = {"same_ren": "Same-Ren", "dif_ren": "Different-Ren"}
BATCH_METHOD_COLORS = {"same_ren": "#2980b9", "dif_ren": "#e67e22"}

BATCH_CSVS = {
    "same_ren": "results_batch_same_ren.csv",
    "dif_ren": "results_batch_dif_ren.csv",
}

def _find_batch_csv(path):
    for candidate in (path, os.path.join("../", path), os.path.join(os.getcwd(), path)):
        if os.path.isfile(candidate):
            return candidate
    return None

_batch_frames = []
for _method, _path in BATCH_CSVS.items():
    _found = _find_batch_csv(_path)
    if _found is None:
        print(f"[batch] {_path} not found yet - skipping '{_method}'")
        continue
    _df = pd.read_csv(_found)
    _df["method"] = _method
    _batch_frames.append(_df)
    print(f"[batch] loaded {len(_df):>4d} rows from {_found} (method={_method})")
```

```
batch_raw = pd.concat(_batch_frames, ignore_index=True) if _batch_frames else
print(f"\nTotal batch rows loaded: {len(batch_raw)}")

if not batch_raw.empty:
    display(batch_raw.groupby(["method", "status"]).size().rename("n_runs").
```

```
[batch] loaded 55 rows from results_batch_same_ren.csv (method=same_ren)
[batch] loaded 42 rows from results_batch_dif_ren.csv (method=dif_ren)
```

Total batch rows loaded: 97

		n_runs
method	status	
dif_ren	ok	13
	timeout_no_solution	29
same_ren	ok	55

```
In [19]: # Clean, de-duplicate (resumed batches append a new row per attempt, so a
# scenario count that first timed out and was later retried has one row per
# attempt) and derive convenience columns used throughout this section.
#
# We keep only the LATEST attempt per (method, num_scenarios_folder) – CSV
# rows are appended in chronological order, so `keep="last"` reflects the
# current state of each scenario-count folder rather than double-counting
# retried folders as both "failed" and "ok".
BATCH_NUMERIC_COLS = [
    "num_scenarios_folder", "num_scenarios_original", "num_scenarios_reduced",
    "preprocess_time_s", "build_instance_time_s", "solve_time_s", "wall_time",
    "num_variables", "num_constraints", "num_binary_vars", "num_integer_vars",
    "lower_bound", "upper_bound", "gurobi_gap_pct",
    "obj_fun", "obj_DA_income", "obj_RM_income", "obj_IM_income",
    "obj_IB_income", "obj_IB_costs", "obj_IB_net", "obj_FD_costs",
    "concurrency_lanes", "threads_per_solve",
]

# batch_run_scenarios.py doesn't record the --time-limit it was invoked
# with, so we track it here manually. Default is the script's own default
# (7200s = 2h); the dif_ren scenario counts below were re-run on 2026-08-22
# with --time-limit 14400 (4h) after timing out at the 2h default.
DEFAULT_TIME_LIMIT_S = 7200
RETRY_TIME_LIMIT_S = 14400
RETRY_4H_SCENARIOS = {
    "dif_ren": {140, 150, 170, 180, 190, 200, 210, 220, 230, 240, 250, 260,
}

def _time_limit_for(row):
    wanted = RETRY_4H_SCENARIOS.get(row["method"], set())
    try:
        folder = int(row["num_scenarios_folder"])
    except (TypeError, ValueError):
        return DEFAULT_TIME_LIMIT_S
    return RETRY_TIME_LIMIT_S if folder in wanted else DEFAULT_TIME_LIMIT_S
```

```

if not batch_raw.empty:
    for _c in BATCH_NUMERIC_COLS:
        if _c in batch_raw.columns:
            batch_raw[_c] = pd.to_numeric(batch_raw[_c], errors="coerce")

    batch_latest = (
        batch_raw
        .drop_duplicates(subset=["method", "num_scenarios_folder"], keep="last")
        .copy()
    )
    batch_latest["time_limit_s"] = batch_latest.apply(_time_limit_for, axis=1)
    n_retried = int((batch_latest["time_limit_s"] == RETRY_TIME_LIMIT_S).sum())
    print(f"Latest attempt per scenario-count folder: {len(batch_latest)} "
          f"(deduplicated from {len(batch_raw)} total attempts logged); "
          f"{n_retried} of those ran with the {RETRY_TIME_LIMIT_S/3600:.0f}h retry limit")

    batch_ok = (
        batch_latest[batch_latest["status"] == "ok"]
        .sort_values("num_scenarios_original")
        .copy()
    )
    batch_ok["method_label"] = batch_ok["method"].map(BATCH_METHOD_LABELS)
    batch_ok["reduction_ratio"] = batch_ok["num_scenarios_reduced"] / batch_ok["num_scenarios_original"]
    batch_ok["solve_time_per_scenario_s"] = batch_ok["solve_time_s"] / batch_ok["num_scenarios_reduced"]
    batch_ok = batch_ok.sort_values(["method", "num_scenarios_original"]).reset_index(drop=True)

    batch_not_ok = batch_latest[batch_latest["status"] != "ok"].copy()
else:
    batch_latest = pd.DataFrame()
    batch_ok = pd.DataFrame()
    batch_not_ok = pd.DataFrame()

print(f"Usable ('ok') runs: {len(batch_ok)} | Non-'ok' runs (latest attempt only): {len(batch_not_ok)}")
if not batch_ok.empty:
    display(batch_ok[["method_label", "num_scenarios_original", "num_scenarios_reduced",
                     "reduction_ratio", "solve_time_s", "obj_fun"]].head(10))

```

Latest attempt per scenario-count folder: 81 (deduplicated from 97 total attempts logged); 16 of those ran with the 4h retry limit.

Usable ('ok') runs: 68 | Non-'ok' runs (latest attempt only): 13

	method_label	num_scenarios_original	num_scenarios_reduced	reduction_rat
0	Different-Ren	50	46	0.920
1	Different-Ren	60	51	0.850
2	Different-Ren	70	55	0.786
3	Different-Ren	80	70	0.875
4	Different-Ren	90	76	0.844
5	Different-Ren	100	88	0.880
6	Different-Ren	110	98	0.891
7	Different-Ren	120	108	0.900
8	Different-Ren	130	108	0.831
9	Different-Ren	140	122	0.871

14.1 · Insight 1 — Time & Resources vs. Scenario Count

Compares wall-clock time (preprocessing, instance build, solve) and model size (variables/constraints) between `same_ren` and `dif_ren`, as a function of both the **original** number of scenarios (before reduction) and the **reduced** number actually solved.

```
In [20]: TIME_COLS = ["preprocess_time_s", "build_instance_time_s", "solve_time_s", "
RESOURCE_COLS = ["num_variables", "num_constraints", "num_binary_vars", "num

if batch_ok.empty:
    print("No batch data available yet - run batch_run_scenarios.py first.")
else:
    print("Mean time components [s] by method and original scenario count:")
    time_table = batch_ok.pivot_table(
        index="num_scenarios_original", columns="method_label", values=TIME_
    ).sort_index()
    display(time_table)

    print("\nMean model-size components by method and original scenario count")
    resource_table = batch_ok.pivot_table(
        index="num_scenarios_original", columns="method_label", values=RESOU
    ).sort_index()
    display(resource_table)
```

Mean time components [s] by method and original scenario count:

method_label	build_instance_time_s		preprocess_time_s		solv	
	Different-Ren	Same-Ren	Different-Ren	Same-Ren	Different-Ren	
num_scenarios_original						
50	1.0624	0.2251	0.5087	0.4131	307.0591	
60	1.2616	0.2373	0.6194	0.4517	798.4808	
70	0.4865	0.4917	0.6280	0.4952	1006.8843	
80	0.6211	1.2439	0.7677	0.6206	929.0547	
90	0.6897	0.8058	0.9166	0.6654	891.8164	
100	0.7797	0.3394	0.9888	0.7156	2113.9994	
110	0.8506	0.8549	1.0315	0.7966	2725.0262	
120	0.9357	0.5510	1.1513	0.9093	4737.8897	
130	0.9487	0.4492	1.2954	0.8964	4642.1270	
140	1.0804	0.5642	1.4052	0.9986	9760.4829	
150	1.1071	0.5389	1.5326	1.0722	6713.4026	
160	1.2000	0.5070	1.5795	1.0984	7276.3032	
170	1.3095	0.5520	1.7508	1.1559	12098.9159	
180	NaN	1.7312	NaN	1.2598	NaN	
190	NaN	0.6574	NaN	1.2823	NaN	
200	NaN	0.6173	NaN	1.3638	NaN	
210	NaN	0.5668	NaN	1.4166	NaN	
220	NaN	0.6901	NaN	1.5547	NaN	
230	NaN	0.7064	NaN	1.5942	NaN	
240	NaN	0.8017	NaN	1.6695	NaN	
250	NaN	0.8230	NaN	1.7279	NaN	
260	NaN	0.9207	NaN	1.8727	NaN	
270	NaN	0.9739	NaN	1.9419	NaN	
280	NaN	0.9336	NaN	1.9633	NaN	
290	NaN	0.9301	NaN	2.0709	NaN	
300	NaN	0.9844	NaN	2.0783	NaN	
310	NaN	0.9184	NaN	2.1535	NaN	
320	NaN	1.1266	NaN	2.2912	NaN	4
330	NaN	1.1598	NaN	2.3108	NaN	2
340	NaN	1.0594	NaN	2.3371	NaN	

method_label	build_instance_time_s		preprocess_time_s		solv	
	Different-Ren	Same-Ren	Different-Ren	Same-Ren	Different-Ren	
num_scenarios_original						
350	NaN	1.2788	NaN	2.5414	NaN	
360	NaN	1.2508	NaN	2.5919	NaN	
370	NaN	1.1109	NaN	2.5332	NaN	
380	NaN	1.1836	NaN	2.6662	NaN	
390	NaN	1.2586	NaN	2.7036	NaN	
400	NaN	1.2807	NaN	2.8279	NaN	
410	NaN	1.2768	NaN	2.8515	NaN	
420	NaN	1.2578	NaN	2.9439	NaN	
430	NaN	1.2759	NaN	2.9809	NaN	
440	NaN	1.3798	NaN	3.0806	NaN	
450	NaN	1.4972	NaN	3.2892	NaN	
460	NaN	1.3186	NaN	3.3611	NaN	
470	NaN	1.4024	NaN	3.2252	NaN	4
480	NaN	1.6094	NaN	3.4106	NaN	1
490	NaN	1.6436	NaN	3.4998	NaN	6
500	NaN	1.6553	NaN	3.6702	NaN	1
510	NaN	1.3832	NaN	3.6847	NaN	
520	NaN	1.6266	NaN	3.6096	NaN	
530	NaN	1.5207	NaN	3.7704	NaN	
540	NaN	1.6631	NaN	3.9652	NaN	
550	NaN	1.6036	NaN	3.7873	NaN	
560	NaN	1.5995	NaN	3.9226	NaN	
570	NaN	1.5624	NaN	4.1323	NaN	
580	NaN	1.6808	NaN	4.0586	NaN	
590	NaN	1.8122	NaN	4.1685	NaN	

Mean model-size components by method and original scenario count:

method_label	num_binary_vars		num_constraints		Di
	Different-Ren	Same-Ren	Different-Ren	Same-Ren	
num_scenarios_original					
50	13248.0000	8352.0000	454089.0000	209518.0000	132
60	14688.0000	8352.0000	534494.0000	216734.0000	146
70	15840.0000	5760.0000	589346.0000	126560.0000	158
80	20160.0000	15552.0000	913643.0000	535896.0000	201
90	21888.0000	14400.0000	937437.0000	472880.0000	218
100	25344.0000	11232.0000	1233539.0000	312770.0000	253
110	28224.0000	15840.0000	1522637.0000	555322.0000	282
120	31104.0000	22176.0000	1751542.0000	926596.0000	311
130	31104.0000	15552.0000	1696651.0000	551932.0000	311
140	35136.0000	21312.0000	2195163.0000	884412.0000	351
150	35424.0000	18432.0000	2196209.0000	678568.0000	354
160	39744.0000	16128.0000	2675257.0000	554380.0000	397
170	42048.0000	18720.0000	2904695.0000	745426.0000	420
180	NaN	20448.0000	NaN	786982.0000	
190	NaN	22752.0000	NaN	970910.0000	
200	NaN	20448.0000	NaN	821566.0000	
210	NaN	16416.0000	NaN	542422.0000	
220	NaN	21888.0000	NaN	870124.0000	
230	NaN	22752.0000	NaN	918222.0000	
240	NaN	26784.0000	NaN	1263654.0000	
250	NaN	26496.0000	NaN	1240616.0000	
260	NaN	31392.0000	NaN	1655258.0000	
270	NaN	33408.0000	NaN	1843812.0000	
280	NaN	31104.0000	NaN	1629304.0000	
290	NaN	30816.0000	NaN	1636510.0000	
300	NaN	32544.0000	NaN	1768206.0000	
310	NaN	28800.0000	NaN	1391116.0000	
320	NaN	39168.0000	NaN	2598992.0000	
330	NaN	40032.0000	NaN	2571830.0000	
340	NaN	33120.0000	NaN	1803158.0000	

method_label	num_binary_vars		num_constraints		Di
	Different-Ren	Same-Ren	Different-Ren	Same-Ren	
num_scenarios_original					
350	NaN	42912.0000	NaN	2866950.0000	
360	NaN	42624.0000	NaN	2839436.0000	
370	NaN	33984.0000	NaN	1868452.0000	
380	NaN	36864.0000	NaN	2222228.0000	
390	NaN	40896.0000	NaN	2669036.0000	
400	NaN	39456.0000	NaN	2423754.0000	
410	NaN	40320.0000	NaN	2602656.0000	
420	NaN	38592.0000	NaN	2362592.0000	
430	NaN	38304.0000	NaN	2337158.0000	
440	NaN	44352.0000	NaN	3071704.0000	
450	NaN	47520.0000	NaN	3443714.0000	
460	NaN	37152.0000	NaN	2174406.0000	
470	NaN	41472.0000	NaN	2736856.0000	
480	NaN	51552.0000	NaN	4012474.0000	
490	NaN	51264.0000	NaN	4026152.0000	
500	NaN	50976.0000	NaN	3914614.0000	
510	NaN	33120.0000	NaN	1800006.0000	
520	NaN	48384.0000	NaN	3653844.0000	
530	NaN	44928.0000	NaN	3091192.0000	
540	NaN	50112.0000	NaN	3735880.0000	
550	NaN	47232.0000	NaN	3449724.0000	
560	NaN	42048.0000	NaN	2755220.0000	
570	NaN	42336.0000	NaN	2761818.0000	
580	NaN	48672.0000	NaN	3520222.0000	
590	NaN	55584.0000	NaN	4575134.0000	

```
In [21]: if not batch_ok.empty:
          fig, axes = plt.subplots(1, 2, figsize=(14, 5))
          fig.suptitle("Fig B1 - Solve Time vs. Scenario Count", fontweight="bold")

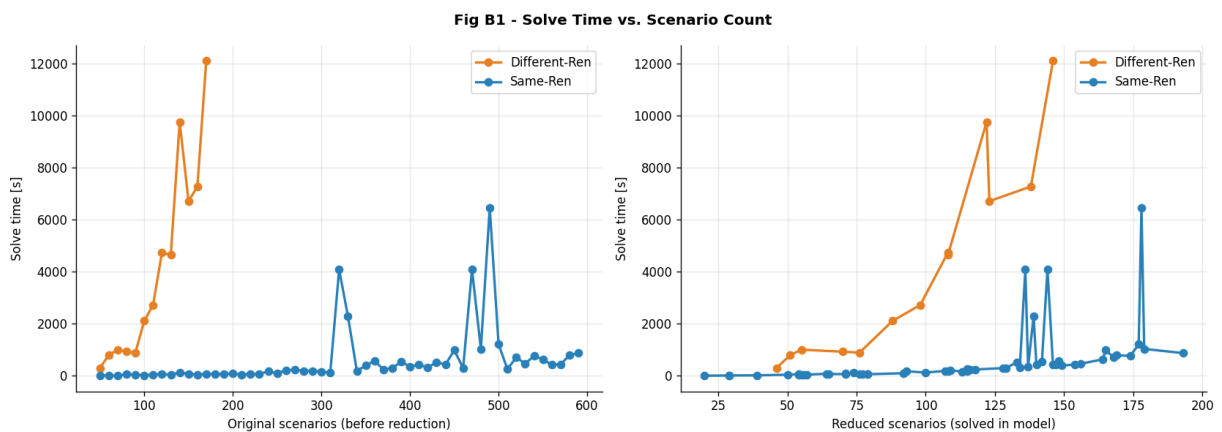
          for ax, xcol, xlabel in zip(
              axes,
              ["num_scenarios_original", "num_scenarios_reduced"],
```

```

        ["Original scenarios (before reduction)", "Reduced scenarios (solved
    ):
    for method, g in batch_ok.groupby("method"):
        g = g.sort_values(xcol)
        ax.plot(g[xcol], g["solve_time_s"], marker="o", lw=2,
                color=BATCH_METHOD_COLORS[method], label=BATCH_METHOD_LA
    ax.set_xlabel(xlabel)
    ax.set_ylabel("Solve time [s]")
    ax.grid(alpha=0.3)
    ax.legend()

    plt.tight_layout()
    _save(fig, "fig_b1_solve_time_vs_scenarios")
    plt.show()
else:
    print("Skipped: no batch data.")

```



```

In [22]: if not batch_ok.empty:
    common_counts = sorted(
        set(batch_ok.loc[batch_ok.method == "same_ren", "num_scenarios_origi
    & set(batch_ok.loc[batch_ok.method == "dif_ren", "num_scenarios_origi
    ])
    print("Original scenario counts completed by BOTH methods so far:", comm

    if common_counts:
        stage_colors = ["#8ecae6", "#219ebc", "#023047"]
        fig, axes = plt.subplots(1, 2, figsize=(14, 5), sharey=True)
        fig.suptitle("Fig B2 - Time Breakdown by Stage (matched scenario cou
        for ax, method in zip(axes, ["same_ren", "dif_ren"]):
            g = (batch_ok[batch_ok.method == method]
                .set_index("num_scenarios_original").reindex(common_counts))
            xpos = np.arange(len(common_counts))
            bottom = np.zeros(len(common_counts))
            for col, color in zip(TIME_COLS[:3], stage_colors):
                vals = g[col].fillna(0).values
                ax.bar(xpos, vals, bottom=bottom, color=color, label=col)
                bottom += vals
            ax.set_xticks(xpos)
            ax.set_xticklabels(common_counts, rotation=45)
            ax.set_title(BATCH_METHOD_LABELS[method])
            ax.set_xlabel("Original scenarios")
            ax.grid(alpha=0.3, axis="y")

```

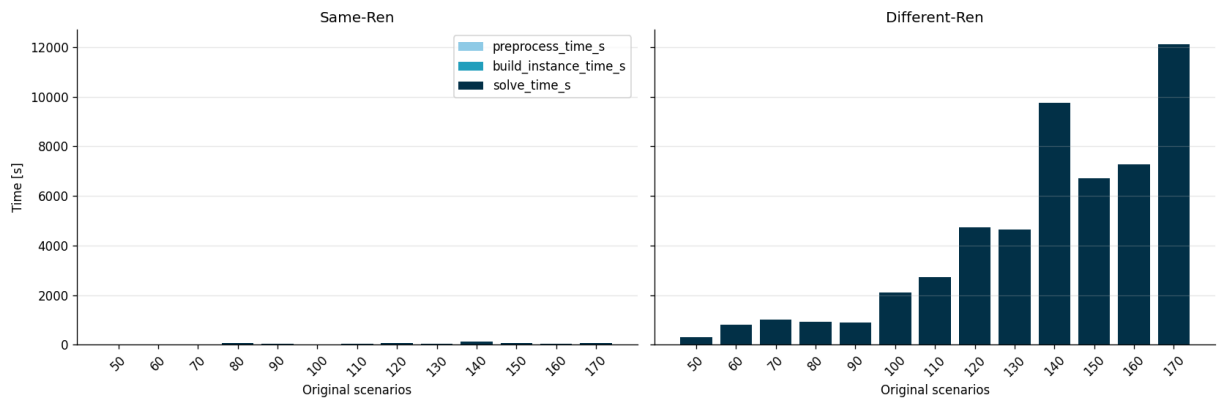
```

        axes[0].set_ylabel("Time [s]")
        axes[0].legend()
        plt.tight_layout()
        _save(fig, "fig_b2_time_breakdown")
        plt.show()
    else:
        print("No overlapping scenario counts between methods yet - re-run 1
else:
    print("Skipped: no batch data.")

```

Original scenario counts completed by BOTH methods so far: [50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170]

Fig B2 - Time Breakdown by Stage (matched scenario counts)

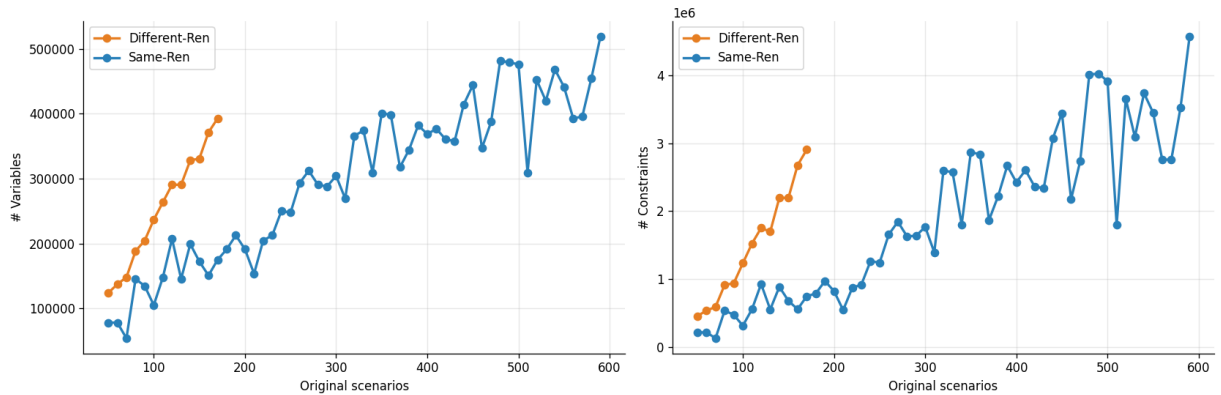


```

In [23]: if not batch_ok.empty:
    fig, axes = plt.subplots(1, 2, figsize=(14, 5))
    fig.suptitle("Fig B3 - Model Size vs. Scenario Count", fontweight="bold")
    for ax, col, ylabel in zip(axes, ["num_variables", "num_constraints"], [
        for method, g in batch_ok.groupby("method"):
            g = g.sort_values("num_scenarios_original")
            ax.plot(g["num_scenarios_original"], g[col], marker="o", lw=2,
                    color=BATCH_METHOD_COLORS[method], label=BATCH_METHOD_LA
            ax.set_xlabel("Original scenarios")
            ax.set_ylabel(ylabel)
            ax.grid(alpha=0.3)
            ax.legend()
        plt.tight_layout()
        _save(fig, "fig_b3_model_size_vs_scenarios")
        plt.show()
    else:
        print("Skipped: no batch data.")

```

Fig B3 - Model Size vs. Scenario Count



14.2 · Insight 2 — Objective Function vs. Scenario Count

Compares the total objective value and its income/cost components between the two methodologies, again as a function of both original and reduced scenario counts.

```
In [24]: OBJ_COLS = ["obj_fun", "obj_DA_income", "obj_RM_income", "obj_IM_income",
                    "obj_IB_income", "obj_IB_costs", "obj_IB_net", "obj_FD_costs"]

if not batch_ok.empty:
    print("Mean objective-function components by method and original scenario count")
    obj_table = batch_ok.pivot_table(
        index="num_scenarios_original", columns="method_label", values=OBJ_COLS
    ).sort_index()
    display(obj_table)
else:
    print("Skipped: no batch data.")
```

Mean objective-function components by method and original scenario count:

method_label	obj_DA_income		obj_FD_costs		ol
	Different-Ren	Same-Ren	Different-Ren	Same-Ren	
num_scenarios_original					
50	-38875.7988	-51571.1304	96.2806	108.3670	610.3474
60	-45442.7365	-48051.9860	94.7126	108.3608	531.2877
70	-39873.7181	-51189.1019	89.3272	111.2036	616.4344
80	-37244.1282	-46506.4061	95.3293	106.7745	619.6909
90	-41246.3705	-53491.1363	90.4249	107.9647	526.7753
100	-33115.3206	-49723.2064	96.8189	107.2424	755.4993
110	-35058.7773	-54072.0088	94.9307	108.4064	652.4287
120	-33048.5766	-57039.4093	92.3255	107.0726	573.2853
130	-39650.6297	-47988.5522	90.6966	105.6167	693.0803
140	-34324.2247	-42899.4013	90.6227	107.7164	652.9322
150	-37604.4554	-51232.5644	92.8088	106.5556	797.5958
160	-34923.9741	-52212.8543	94.1233	104.0768	531.5415
170	-37132.6023	-51101.3936	92.7390	109.0839	693.0813
180	NaN	-51604.0281	NaN	104.9465	NaN
190	NaN	-52168.2006	NaN	108.8245	NaN
200	NaN	-46569.4387	NaN	108.9549	NaN
210	NaN	-44185.8530	NaN	106.8108	NaN
220	NaN	-47578.8511	NaN	106.6871	NaN
230	NaN	-53576.4745	NaN	107.6604	NaN
240	NaN	-51386.3943	NaN	106.5269	NaN
250	NaN	-56931.4470	NaN	107.5228	NaN
260	NaN	-50092.1658	NaN	109.6838	NaN
270	NaN	-54768.1972	NaN	108.4486	NaN
280	NaN	-48413.2033	NaN	108.6142	NaN
290	NaN	-50399.6913	NaN	106.0226	NaN
300	NaN	-49143.9334	NaN	105.0098	NaN
310	NaN	-48723.4686	NaN	106.2137	NaN
320	NaN	-48810.4290	NaN	106.4252	NaN
330	NaN	-50782.0947	NaN	107.1300	NaN
340	NaN	-49824.8775	NaN	106.0810	NaN

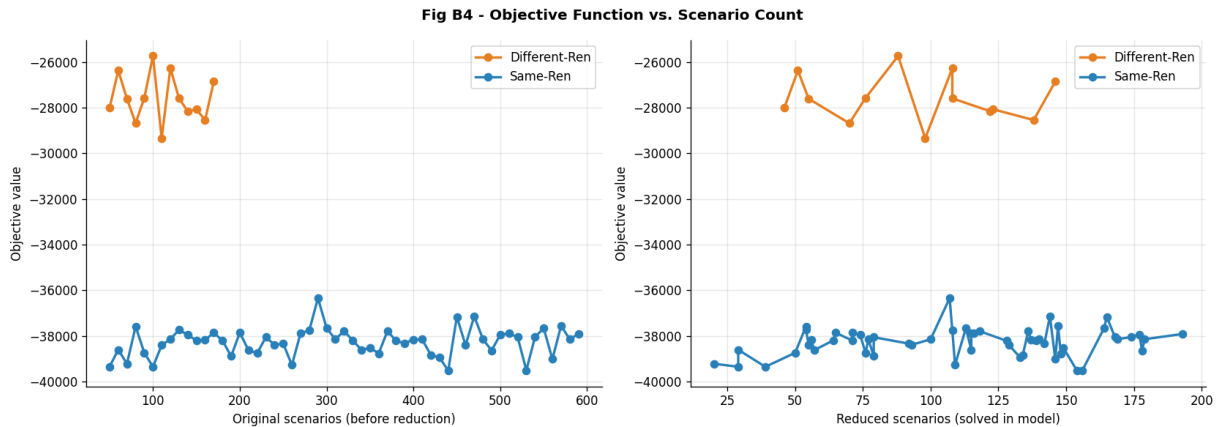
method_label	obj_DA_income		obj_FD_costs		ol
	Different-Ren	Same-Ren	Different-Ren	Same-Ren	
num_scenarios_original					
350	NaN	-50974.9351	NaN	105.8354	NaN
360	NaN	-47892.3473	NaN	107.9800	NaN
370	NaN	-49471.7689	NaN	106.5742	NaN
380	NaN	-48797.3371	NaN	107.0497	NaN
390	NaN	-48282.9295	NaN	107.1220	NaN
400	NaN	-50301.2650	NaN	105.1122	NaN
410	NaN	-46765.5107	NaN	109.2980	NaN
420	NaN	-45995.2445	NaN	107.6906	NaN
430	NaN	-50445.2594	NaN	104.8371	NaN
440	NaN	-52076.9024	NaN	108.6536	NaN
450	NaN	-48261.3558	NaN	105.2083	NaN
460	NaN	-49270.5712	NaN	107.5095	NaN
470	NaN	-47566.9593	NaN	106.5317	NaN
480	NaN	-51046.2703	NaN	106.6604	NaN
490	NaN	-49948.7678	NaN	106.9228	NaN
500	NaN	-50657.0508	NaN	105.4008	NaN
510	NaN	-48440.7708	NaN	104.8384	NaN
520	NaN	-49663.3451	NaN	108.4241	NaN
530	NaN	-48876.0076	NaN	106.9612	NaN
540	NaN	-49815.8434	NaN	107.5076	NaN
550	NaN	-47530.4864	NaN	107.1777	NaN
560	NaN	-48436.6593	NaN	106.1726	NaN
570	NaN	-50500.8763	NaN	106.1164	NaN
580	NaN	-47761.0597	NaN	106.7478	NaN
590	NaN	-57154.5421	NaN	107.9080	NaN

```
In [25]: if not batch_ok.empty:
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
fig.suptitle("Fig B4 - Objective Function vs. Scenario Count", fontweight
for ax, xcol, xlabel in zip(
    axes,
    ["num_scenarios_original", "num_scenarios_reduced"],
    ["Original scenarios (before reduction)", "Reduced scenarios (solved
```

```

):
    for method, g in batch_ok.groupby("method"):
        g = g.sort_values(xcol)
        ax.plot(g[xcol], g["obj_fun"], marker="o", lw=2,
                color=BATCH_METHOD_COLORS[method], label=BATCH_METHOD_LA
        ax.set_xlabel(xlabel)
        ax.set_ylabel("Objective value")
        ax.grid(alpha=0.3)
        ax.legend()
    plt.tight_layout()
    _save(fig, "fig_b4_objective_vs_scenarios")
    plt.show()
else:
    print("Skipped: no batch data.")

```



```

In [26]: if not batch_ok.empty and common_counts:
    income_cols = ["obj_DA_income", "obj_RM_income", "obj_IM_income", "obj_I
    cost_cols = ["obj_IB_costs", "obj_FD_costs"]
    income_colors = ["#2a9d8f", "#8ab17d", "#e9c46a", "#f4a261"]
    cost_colors = ["#e76f51", "#6a4c93"]

    fig, axes = plt.subplots(1, 2, figsize=(14, 5), sharey=True)
    fig.suptitle("Fig B5 - Objective Components by Method (matched scenario
    for ax, method in zip(axes, ["same_ren", "dif_ren"]):
        g = (batch_ok[batch_ok.method == method]
              .set_index("num_scenarios_original").reindex(common_counts))
        xpos = np.arange(len(common_counts))

        bottom = np.zeros(len(common_counts))
        for col, color in zip(income_cols, income_colors):
            vals = g[col].fillna(0).values
            ax.bar(xpos, vals, bottom=bottom, color=color, label=col)
            bottom += vals

        bottom = np.zeros(len(common_counts))
        for col, color in zip(cost_cols, cost_colors):
            vals = -g[col].fillna(0).values # costs plotted as negative bar
            ax.bar(xpos, vals, bottom=bottom, color=color, label=col, hatch=
            bottom += vals

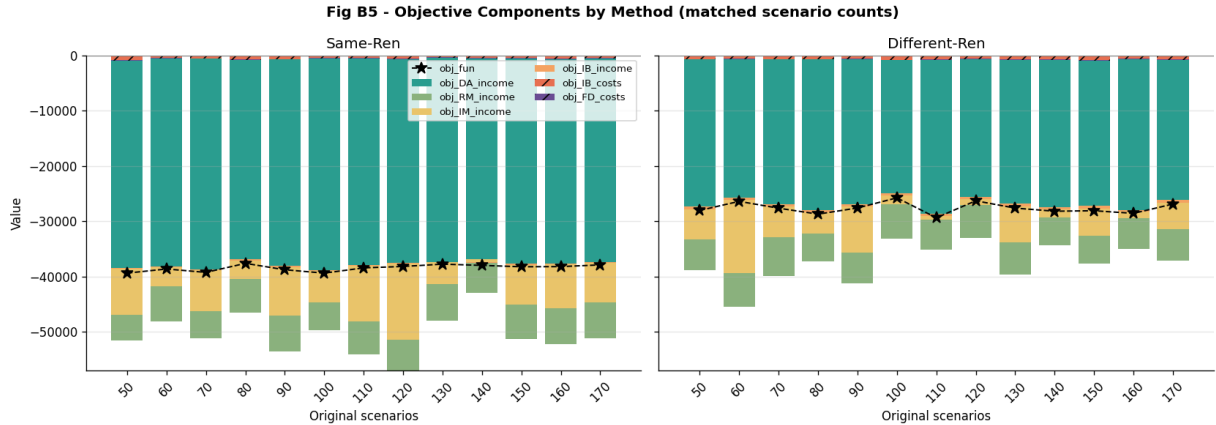
        ax.plot(xpos, g["obj_fun"].values, "k*--", lw=1.2, ms=10, label="obj
        ax.axhline(0, color="grey", lw=0.8)

```

```

        ax.set_xticks(xpos)
        ax.set_xticklabels(common_counts, rotation=45)
        ax.set_title(BATCH_METHOD_LABELS[method])
        ax.set_xlabel("Original scenarios")
        ax.grid(alpha=0.3, axis="y")
        axes[0].set_ylabel("Value")
        axes[0].legend(fontsize=8, ncol=2)
        plt.tight_layout()
        _save(fig, "fig_b5_objective_components")
        plt.show()
    else:
        print("Skipped: no batch data or no overlapping scenario counts yet.")

```

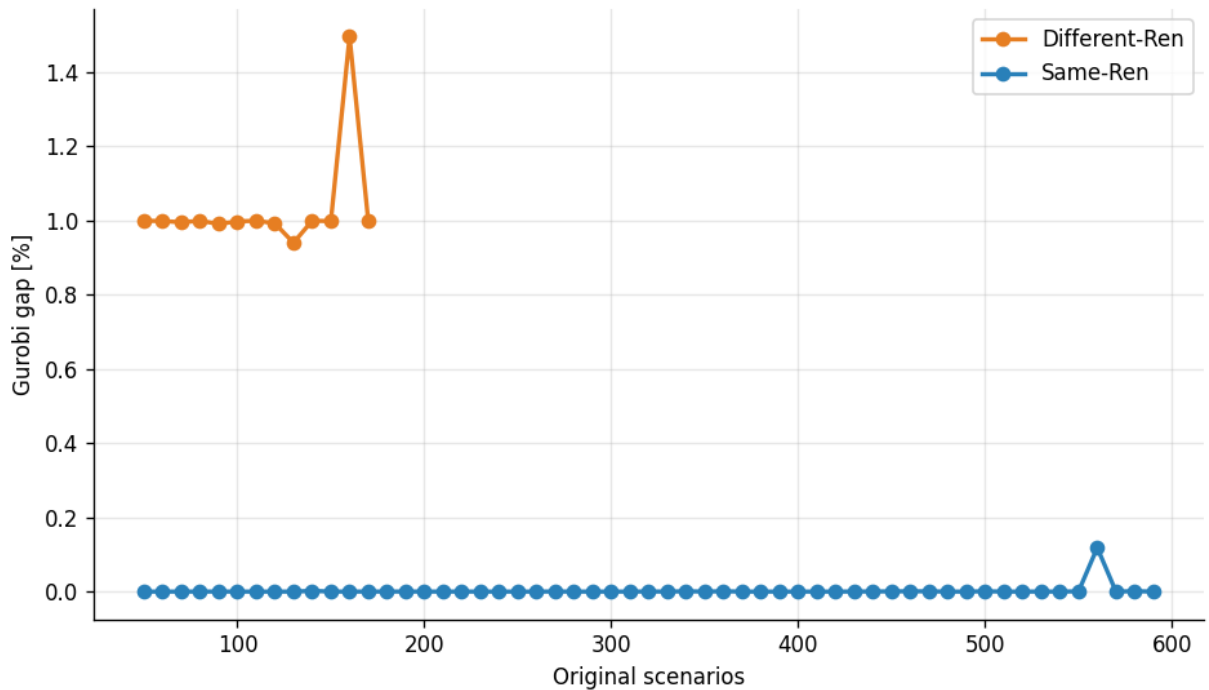


```

In [27]: if not batch_ok.empty and batch_ok["gurobi_gap_pct"].notna().any():
        fig, ax = plt.subplots(figsize=(8, 5))
        fig.suptitle("Fig B6 - Optimality Gap vs. Scenario Count", fontweight="b")
        for method, g in batch_ok.groupby("method"):
            g = g.sort_values("num_scenarios_original")
            ax.plot(g["num_scenarios_original"], g["gurobi_gap_pct"], marker="o",
                    color=BATCH_METHOD_COLORS[method], label=BATCH_METHOD_LABELS[method])
        ax.set_xlabel("Original scenarios")
        ax.set_ylabel("Gurobi gap [%]")
        ax.grid(alpha=0.3)
        ax.legend()
        plt.tight_layout()
        _save(fig, "fig_b6_optimality_gap")
        plt.show()
    else:
        print("Skipped: no gap data available.")

```

Fig B6 - Optimality Gap vs. Scenario Count



14.3 · Insight 3 — Day-Ahead Commitment Comparison

Compares the actual DA commitments (`eDA_p` = sold, `eDA_m` = bought) produced by each methodology, matched at the **same** (or closest) original scenario count so the comparison isolates the effect of the reduction method rather than scenario count.

Each run's `results_dir` (from the batch CSV, e.g. `.../DA/ec/001`) is converted to its sibling market-results folder (`.../DA/market/001/DA/eDA_p.txt`) where the per-scenario, per-period commitments are stored as one row per (reduced) scenario: `scenario_id` probability `v_t=1 ... v_t=T`. Profiles below are probability-weighted means across scenarios.

```
In [28]: def _da_market_dir(results_dir):
# '.../<famscen>/DA/ec/<sim>' -> '.../<famscen>/DA/market/<sim>' (see si
results_dir = str(results_dir)
marker = os.sep + "ec" + os.sep
if marker not in results_dir:
    return None
return results_dir.replace(marker, os.sep + "market" + os.sep)

def load_da_commitment(results_dir, filename):
# eDA_p.txt / eDA_m.txt: rows = reduced scenarios, cols = [id, prob, v_t
market_dir = _da_market_dir(results_dir)
if market_dir is None:
    return None
```

```

path = os.path.join(market_dir, "DA", filename)
if not os.path.isfile(path):
    return None
df = pd.read_csv(path, sep=r"\s+", header=None, engine="python")
if df.shape[1] < 3:
    return None
probs = df.iloc[:, 1].to_numpy(dtype=float)
values = df.iloc[:, 2:].to_numpy(dtype=float)
if probs.sum() > 0 and not np.isclose(probs.sum(), 1.0, atol=1e-3):
    probs = probs / probs.sum()
return probs, values # probs: (S,), values: (S, T)

def weighted_commitment_profile(results_dir):
    # Probability-weighted mean eDA_p, eDA_m and net (sold - bought) profile
    p = load_da_commitment(results_dir, "eDA_p.txt")
    m = load_da_commitment(results_dir, "eDA_m.txt")
    if p is None or m is None:
        return None
    probs_p, vals_p = p
    probs_m, vals_m = m
    mean_p = (probs_p[:, None] * vals_p).sum(axis=0)
    mean_m = (probs_m[:, None] * vals_m).sum(axis=0)
    return mean_p, mean_m, mean_p - mean_m

print("Commitment-loading helpers ready.")

```

Commitment-loading helpers ready.

```

In [29]: commit_records = []
if not batch_ok.empty:
    for _, row in batch_ok.iterrows():
        prof = weighted_commitment_profile(row["results_dir"])
        if prof is None:
            continue
        mean_p, mean_m, net = prof
        dt_h = 24.0 / len(mean_p) # hours per period (e.g. 0.25 for 15-min,
        commit_records.append({
            "method": row["method"],
            "num_scenarios_original": row["num_scenarios_original"],
            "num_scenarios_reduced": row["num_scenarios_reduced"],
            "results_dir": row["results_dir"],
            "T_periods": len(mean_p),
            "total_sold_MWh": mean_p.sum() * dt_h,
            "total_bought_MWh": mean_m.sum() * dt_h,
            "net_MWh": net.sum() * dt_h,
            "peak_sold_MW": mean_p.max(),
            "peak_bought_MW": mean_m.max(),
        })

commit_df = pd.DataFrame(commit_records)
if not commit_df.empty:
    commit_df["method_label"] = commit_df["method"].map(BATCH_METHOD_LABELS)
    commit_df = commit_df.sort_values(["method", "num_scenarios_original"])

print(f"Loaded commitment profiles for {len(commit_df)} / {len(batch_ok)} 'c

```

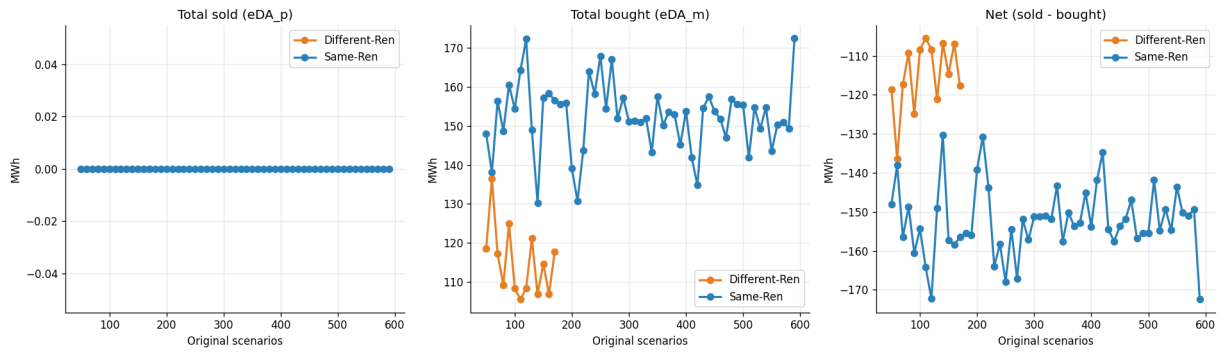
```
if not commit_df.empty:
    display(commit_df.drop(columns=["results_dir"]).head(10))
```

Loaded commitment profiles for 68 / 68 'ok' runs.

	method	num_scenarios_original	num_scenarios_reduced	T_periods	total_so
0	dif_ren	50	46	96	
1	dif_ren	60	51	96	
2	dif_ren	70	55	96	
3	dif_ren	80	70	96	
4	dif_ren	90	76	96	
5	dif_ren	100	88	96	
6	dif_ren	110	98	96	
7	dif_ren	120	108	96	
8	dif_ren	130	108	96	
9	dif_ren	140	122	96	

```
In [30]: if not commit_df.empty:
fig, axes = plt.subplots(1, 3, figsize=(16, 5))
fig.suptitle("Fig B7 - Day-Ahead Commitment Totals vs. Scenario Count",
for ax, col, title in zip(
    axes,
    ["total_sold_MWh", "total_bought_MWh", "net_MWh"],
    ["Total sold (eDA_p)", "Total bought (eDA_m)", "Net (sold - bought)"
]):
    for method, g in commit_df.groupby("method"):
        g = g.sort_values("num_scenarios_original")
        ax.plot(g["num_scenarios_original"], g[col], marker="o", lw=2,
                color=BATCH_METHOD_COLORS[method], label=BATCH_METHOD_LA
ax.set_title(title)
ax.set_xlabel("Original scenarios")
ax.set_ylabel("MWh")
ax.grid(alpha=0.3)
ax.legend()
plt.tight_layout()
_save(fig, "fig_b7_commitment_totals")
plt.show()
else:
    print("Skipped: no commitment data available yet (result files not gener
```

Fig B7 - Day-Ahead Commitment Totals vs. Scenario Count



```
In [31]: if not commit_df.empty:
    matched_counts = sorted(
        set(commit_df.loc[commit_df.method == "same_ren", "num_scenarios_ori"]
            & set(commit_df.loc[commit_df.method == "dif_ren", "num_scenarios_ori"]
        )
    )
    show_counts = matched_counts
    if len(matched_counts) > 6:
        idx = np.linspace(0, len(matched_counts) - 1, 6).round().astype(int)
        show_counts = [matched_counts[i] for i in sorted(set(idx))]
    print("Scenario counts shown (matched between methods):", show_counts)

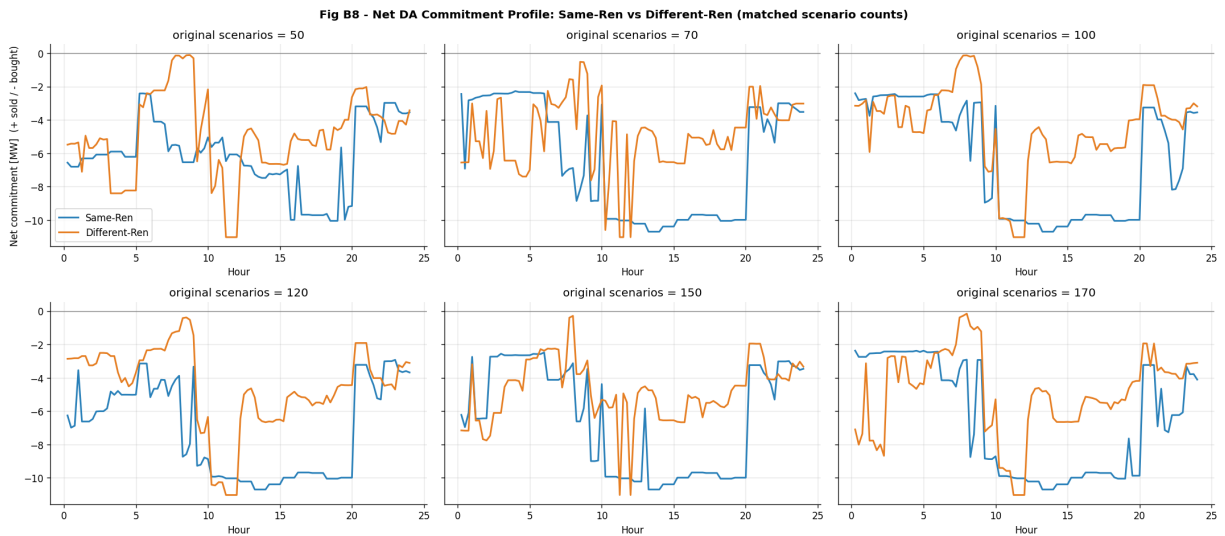
    if show_counts:
        ncols = min(3, len(show_counts))
        nrows = int(np.ceil(len(show_counts) / ncols))
        fig, axes = plt.subplots(nrows, ncols, figsize=(6 * ncols, 4 * nrows))
        axes = axes.flatten()
        fig.suptitle("Fig B8 - Net DA Commitment Profile: Same-Ren vs Differ",
                     fontweight="bold")
        for ax, sc in zip(axes, show_counts):
            for method in ["same_ren", "dif_ren"]:
                r = batch_ok[(batch_ok.method == method) & (batch_ok.num_scenarios == sc)]
                if r.empty:
                    continue
                prof = weighted_commitment_profile(r.iloc[0]["results_dir"])
                if prof is None:
                    continue
                _, _, net = prof
                t = np.arange(1, len(net) + 1) * (24.0 / len(net))
                ax.plot(t, net, lw=1.8, color=BATCH_METHOD_COLORS[method], label=method)
                ax.axhline(0, color="grey", lw=0.8)
                ax.set_title(f"original scenarios = {sc}")
                ax.set_xlabel("Hour")
                ax.grid(alpha=0.3)
            axes[0].set_ylabel("Net commitment [MW] (+ sold / - bought)")
            axes[0].legend()
            for ax in axes[len(show_counts):]:
                ax.axis("off")
        plt.tight_layout()
        _save(fig, "fig_b8_commitment_profiles")
        plt.show()
    else:
        print("No scenario count is present for both methods yet - re-run la")
```

```

else:
    print("Skipped: no commitment data.")

```

Scenario counts shown (matched between methods): [50, 70, 100, 120, 150, 170]



```

In [32]: if not commit_df.empty and matched_counts:
    diffs = []
    for sc in matched_counts:
        profiles = {}
        for method in ["same_ren", "dif_ren"]:
            r = batch_ok[(batch_ok.method == method) & (batch_ok.num_scenari
            if r.empty:
                continue
            prof = weighted_commitment_profile(r.iloc[0]["results_dir"])
            if prof is None:
                continue
            profiles[method] = prof[2] # net profile
        if "same_ren" in profiles and "dif_ren" in profiles and \
            len(profiles["same_ren"]) == len(profiles["dif_ren"]):
            mad = float(np.mean(np.abs(profiles["same_ren"] - profiles["dif
            diffs.append({"num_scenarios_original": sc, "mean_abs_diff_MW":

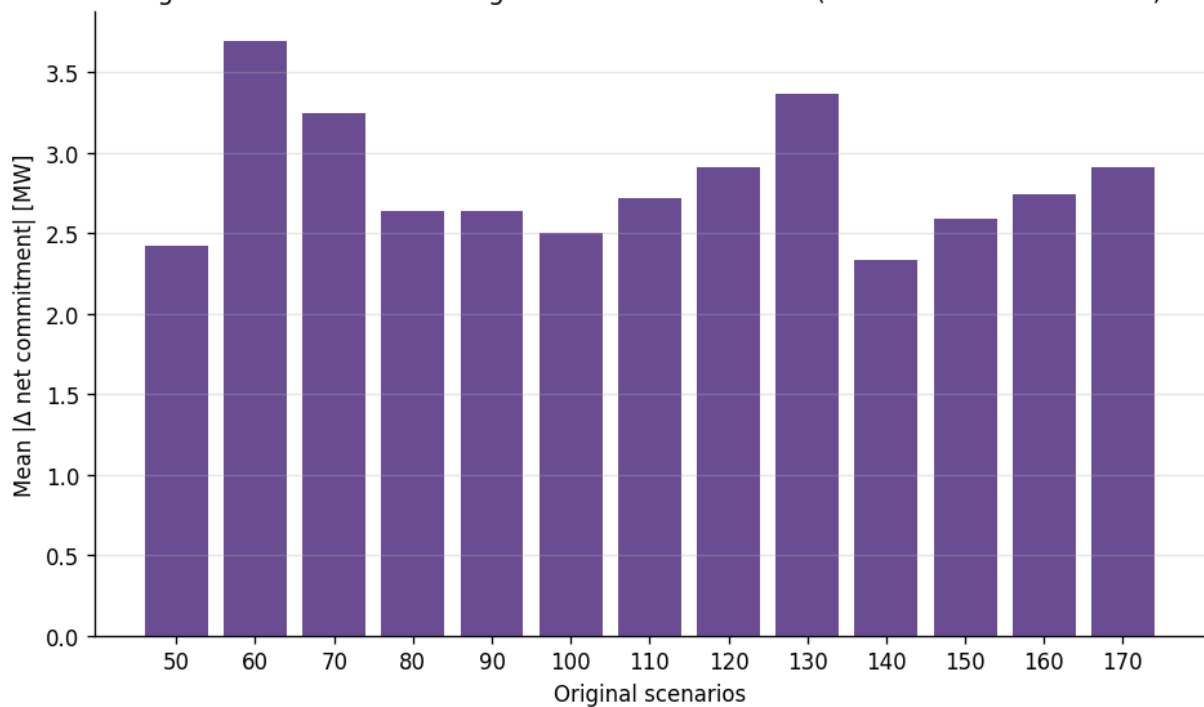
    diff_df = pd.DataFrame(diffs)
    display(diff_df)

    if not diff_df.empty:
        fig, ax = plt.subplots(figsize=(8, 5))
        ax.bar(diff_df["num_scenarios_original"].astype(str), diff_df["mean_
        ax.set_xlabel("Original scenarios")
        ax.set_ylabel("Mean |Δ net commitment| [MW]")
        ax.set_title("Fig B9 - Commitment Divergence Between Methods (matche
        ax.grid(alpha=0.3, axis="y")
        plt.tight_layout()
        _save(fig, "fig_b9_commitment_divergence")
        plt.show()
    else:
        print("Skipped: no matched scenario counts with commitment data yet.")

```

	num_scenarios_original	mean_abs_diff_MW
0	50	2.4230
1	60	3.6945
2	70	3.2495
3	80	2.6425
4	90	2.6424
5	100	2.5007
6	110	2.7179
7	120	2.9083
8	130	3.3640
9	140	2.3352
10	150	2.5933
11	160	2.7393
12	170	2.9137

Fig B9 - Commitment Divergence Between Methods (matched scenario counts)



14.4 · Additional Metrics

A few extra diagnostics that are useful when judging which reduction method is "better" beyond the three requested comparisons:

- **Reduction ratio** — how aggressively each method prunes the original scenario tree.

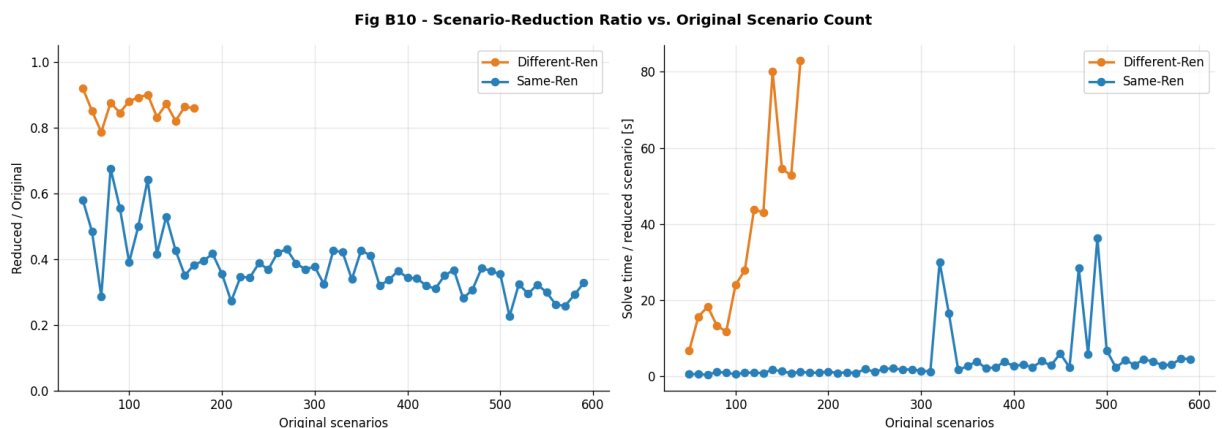
- **Solve time per (reduced) scenario** — a size-normalized efficiency metric, since `dif_ren` and `same_ren` can keep a different number of scenarios for the same original count.
- **Run-status summary** — surfaces any `error` / `timeout_no_solution` rows so incomplete batches are visible at a glance.

```
In [33]: if not batch_ok.empty:
    fig, axes = plt.subplots(1, 2, figsize=(14, 5))
    fig.suptitle("Fig B10 - Scenario-Reduction Ratio vs. Original Scenario C

    ax = axes[0]
    for method, g in batch_ok.groupby("method"):
        g = g.sort_values("num_scenarios_original")
        ax.plot(g["num_scenarios_original"], g["reduction_ratio"], marker="c",
                color=BATCH_METHOD_COLORS[method], label=BATCH_METHOD_LABELS[method])
    ax.set_xlabel("Original scenarios")
    ax.set_ylabel("Reduced / Original")
    ax.set_ylim(0, 1.05)
    ax.grid(alpha=0.3)
    ax.legend()

    ax = axes[1]
    for method, g in batch_ok.groupby("method"):
        g = g.sort_values("num_scenarios_original")
        ax.plot(g["num_scenarios_original"], g["solve_time_per_scenario_s"],
                color=BATCH_METHOD_COLORS[method], label=BATCH_METHOD_LABELS[method])
    ax.set_xlabel("Original scenarios")
    ax.set_ylabel("Solve time / reduced scenario [s]")
    ax.grid(alpha=0.3)
    ax.legend()

    plt.tight_layout()
    _save(fig, "fig_b10_reduction_ratio_and_efficiency")
    plt.show()
else:
    print("Skipped: no batch data.")
```



```
In [34]: if not batch_latest.empty:
    print("Run-status summary by method (latest attempt per scenario-count f
    display(batch_latest.groupby(["method", "status"]).size().rename("n_runs
```

```

if not batch_not_ok.empty:
    print("\nNon-'ok' runs - latest attempt only (inspect 'error' for de
    display(batch_not_ok[["method", "num_scenarios_folder", "status",
                          "time_limit_s", "solve_time_s", "error"]]
            .sort_values(["method", "num_scenarios_folder"]))
else:
    print("\nAll loaded scenario-count folders completed with status 'ok
else:
    print("Skipped: no batch data.")

```

Run-status summary by method (latest attempt per scenario-count folder):

n_runs		
method	status	
dif_ren	ok	13
	timeout_no_solution	13
same_ren	ok	55

Non-'ok' runs - latest attempt only (inspect 'error' for details):

	method	num_scenarios_folder	status	time_limit_s	solve_time_s
84	dif_ren	180	timeout_no_solution	14400	14495.5853
85	dif_ren	190	timeout_no_solution	14400	14506.0084
86	dif_ren	200	timeout_no_solution	14400	14510.0504
88	dif_ren	210	timeout_no_solution	14400	14520.8541
87	dif_ren	220	timeout_no_solution	14400	14514.1444
89	dif_ren	230	timeout_no_solution	14400	14534.1790
90	dif_ren	240	timeout_no_solution	14400	14551.6284
93	dif_ren	250	timeout_no_solution	14400	14590.2904
92	dif_ren	260	timeout_no_solution	14400	14578.3640
91	dif_ren	270	timeout_no_solution	14400	14562.0996
94	dif_ren	280	timeout_no_solution	14400	14585.4491

	method	num_scenarios_folder	status	time_limit_s	solve_time_s
95	dif_ren	290	timeout_no_solution	14400	14586.1917
96	dif_ren	300	timeout_no_solution	14400	14625.0884

14.5 · Export Batch Summary Tables

```
In [35]: if CONFIG["save_figs"]:
    os.makedirs(CONFIG["fig_dir"], exist_ok=True)
    if not batch_ok.empty:
        out1 = os.path.join(CONFIG["fig_dir"], "batch_summary.csv")
        batch_ok.drop(columns=[c for c in ["traceback"] if c in batch_ok.columns])
        print(f"Saved -> {out1}")
    if not commit_df.empty:
        out2 = os.path.join(CONFIG["fig_dir"], "commitment_summary.csv")
        commit_df.drop(columns=["results_dir"]).to_csv(out2, index=False)
        print(f"Saved -> {out2}")
    else:
        print("Set CONFIG['save_figs'] = True to export batch_summary.csv / commitment_summary.csv.")
```

Set CONFIG['save_figs'] = True to export batch_summary.csv / commitment_summary.csv.